The Campus of Tomorrow

Higher Colleges of Technology

كليات التقنية العليا

CDS 2413- PROGRAMMING FOR DATA SCIENCE

Week 4

Week 4	CLO 2: Apply data analysis and visualization techniques to support data-driven decision-making

Document Revision CONTROL (DRC)

Version	Author	Effective Date	Change Description	DRC No
1.0	Mohammed Abdul Rahim	2/05/2023	Define the first version	001

X.Y
X: Major change
Y: minor change

Lecture Outline

- Visualization
- Exploratory Analysis
- Different Charts/Graphs
- Python Programming for Charts/Graphs

LECTURE OBJECTIVES

- The student should be able to do the following at the end of the lecture:
  - Understand the purpose of visualizing the data
  - Investigate the concept of exploratory analysis in data science field
  - Explore the types of visualization(Graphs/Charts)
  - Able to examine the visualization
  - Learn how to write python programming for visualization

What is Exploratory Analysis?

- Exploratory Data Analysis is a method of Data Analysis that employs a variety of techniques to:
  - Gain intuition about the data
  - Perform sanity tests (To ensure that the dataset from which we are deriving conclusions is the correct one).
  - Determine where data is missing.
  - Determine whether any outliers exist.
  - Summarize the data.

Exploratory Analysis

- Steps in Exploratory Analysis

Exploratory Analysis

- **Variable Identification**
  - Identify each variable and its type using commands like info() or dtypes.
  - Alter data types as needed using astype().

1

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23 entries, 0 to 22
Data columns (total 6 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   ID         23 non-null     object
 1   Division   23 non-null     object
 2   Quiz       22 non-null     float64
 3   Practical  23 non-null     int64
 4   Project    23 non-null     float64
 5   Final      23 non-null     int64
dtypes: float64(2), int64(2), object(2)
memory usage: 1.2+ KB
```

2

```
df.dtypes
```

```
ID             object
Division       object
Quiz          float64
Practical       int64
Project       float64
Final           int64
dtype: object
```

Final column data type is int

3 Converting Final column data type from int to float

```
df['Final'] = df['Final'].astype(float)
df.dtypes
```

```
ID             object
Division       object
Quiz          float64
Practical       int64
Project       float64
Final         float64
dtype: object
```

Exploratory Analysis

- **Univariate Analysis**
- In Univariate Analysis, the single variable(column) in the dataset are investigated.
- There are two varieties of variables to be analysed:
  - continuous(decimal) and
  - categorical(Male/Female).
- Following is a reference sheet of various graphical techniques that can be applied to their analysis.

Type Of Variable	Graphical Techniques
Continuous Variable	Histogram, KDE, Boxplot and Q-Q plot(specifically for outliers)
Categorical Variable	Bar Plot, Pie Chart, Frequency Table

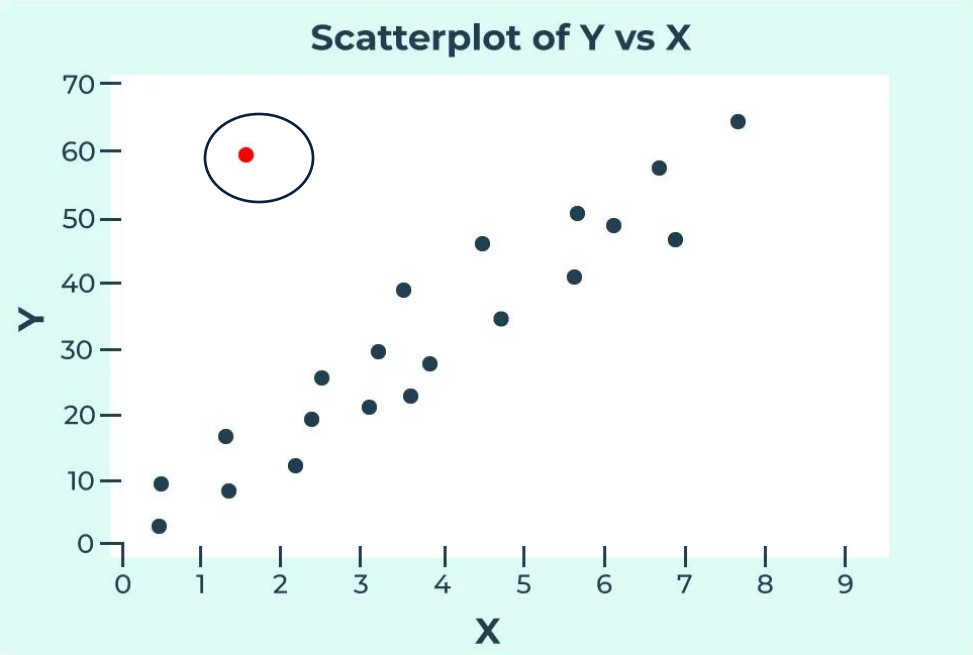
Exploratory Analysis

- **Bivariate Analysis**
- Bivariate Analysis examines the relationship between any two variables, which may be :
  - categorical-continuous,
  - categorical-categorical,
  - continuous-continuous

Type Of Relationship	Graphical Techniques
Continuous-Continuous	Scatter plot, HeatMap, JointPlot, PairPlot.
Categorical-Contnuous	Factor plot, SwarmMap, ViolinPlot, StripPlot.
Categorical-Categorical	Crosstab, Stacked Bar, Barchart.

Exploratory Analysis

- **Missing Value Treatment :**
  - Identifying the missing value and treating them
    - Some of the ways to treat the missing values are finding the mean(average) and filling the missing value with average
    - If missing value is not treated, it will affect the model performance
- **Outlier Removal :**
  - Outlier are the data points which are far away from the other data points
  - Outlier should be identified graphically or other statistical method and **removed from the dataset**



Exploratory Analysis - Quiz

- Provide an example of univariate, bivariate, and multivariate analysis for the "Marks" dataset.
- Include graphs/charts to analyze outliers using exploratory data analysis.

ID	Division	Quiz	Practical	Project	Final
H00247	EDU	84	67	69.6	67
H00258	HLT	74	68	67	73
H00269	ENG	88	86	75.6	74
H00224	EDU		71	65.8	71
H00269	EDU	22	65	66.8	78
H00277	ENG	40	68	76.4	68
H00277	HLT	71	77	64.6	70
H00293	BUS	50	74	60.6	78
H00247	CIS	61	70	66.4	82
H00149	APP	97	98	85.2	87
H00295	EDU	70	76	81	84
H00284	CIS	48	64	81.4	68
H00307	ENG	67	77	73	83
H00293	APP	38	89	64.6	77
H00293	BUS	80	76	68.4	83
H00293	APP	83	62	77	85
H00248	BUS	73	86	76.4	77

Exploratory Analysis - Quiz

Analysis Type	Purpose	Methods	Example
Univariate	Examine the distribution of each variable individually.	Descriptive Statistics, Histograms, Box Plots	Calculate the mean, median, standard deviation, etc., of the "Quiz" scores. Create a histogram to visualize the distribution and identify potential outliers. A box plot can also be used to visually identify outliers.
Bivariate	Investigate the relationship between two variables.	Scatter Plots, Correlation Coefficient, Cross-tabulation	Create a scatter plot to visualize the relationship between "Quiz" scores and "Final" scores. Calculate the correlation coefficient to quantify the strength of this relationship.
Multivariate	Examine the relationship between multiple variables simultaneously.	Multiple Regression, Principal Component Analysis (PCA)	Perform multiple regression to predict "Final" scores using "Quiz", "Practical", and "Project" as predictors.

What is Data Visualization?

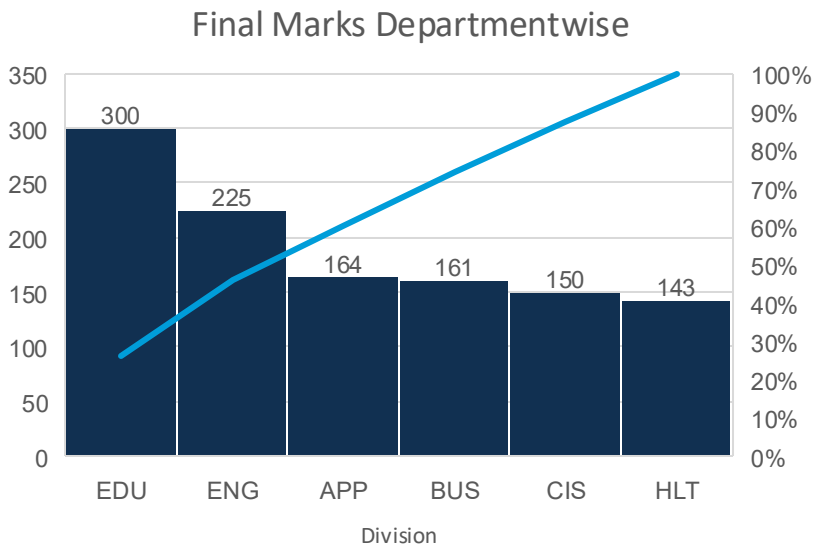
- Data visualization is the visual representation of data and information.
- Facilitate the identification and comprehension of trends, outliers, and patterns in data.
- An excellent method to present information to non-technical audiences without causing confusion.

What is Data Visualization?

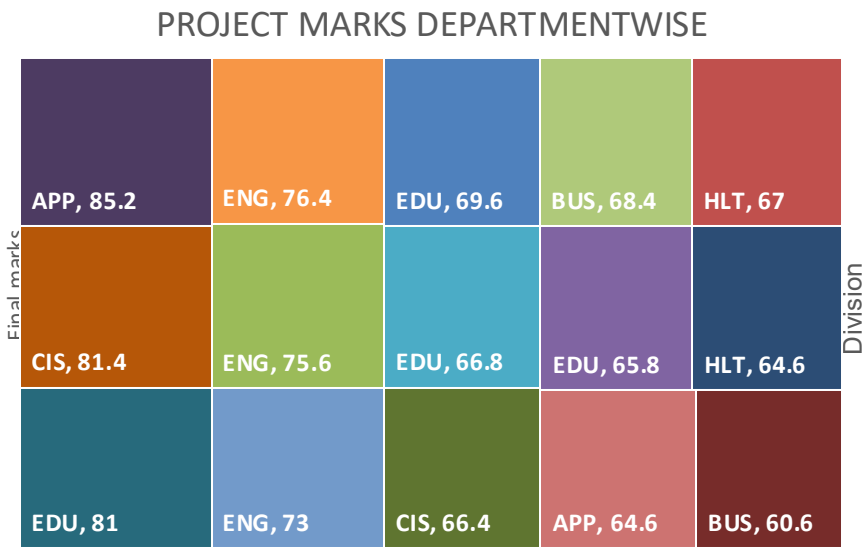
Which division performed best on the quiz, project, or final exam, in your opinion?

ID	Division	Quiz	Practical	Project	Final
H00247	EDU	84	67	69.6	67
H00258	HLT	74	68	67	73
H00269	ENG	88	86	75.6	74
H00224	EDU	56	71	65.8	71
H00269	EDU	22	65	66.8	78
H00277	ENG	40	68	76.4	68
H00277	HLT	71	75	64.6	70
H00293	BUS	50	74	60.6	78
H00247	CIS	61	70	66.4	82
H00149	APP	97	98	85.2	87
H00295	EDU	70	76	81	84
H00284	CIS	48	64	81.4	68
H00307	ENG	67	77	73	83
H00293	APP	38	62	64.6	77
H00293	BUS	80	76	68.4	83

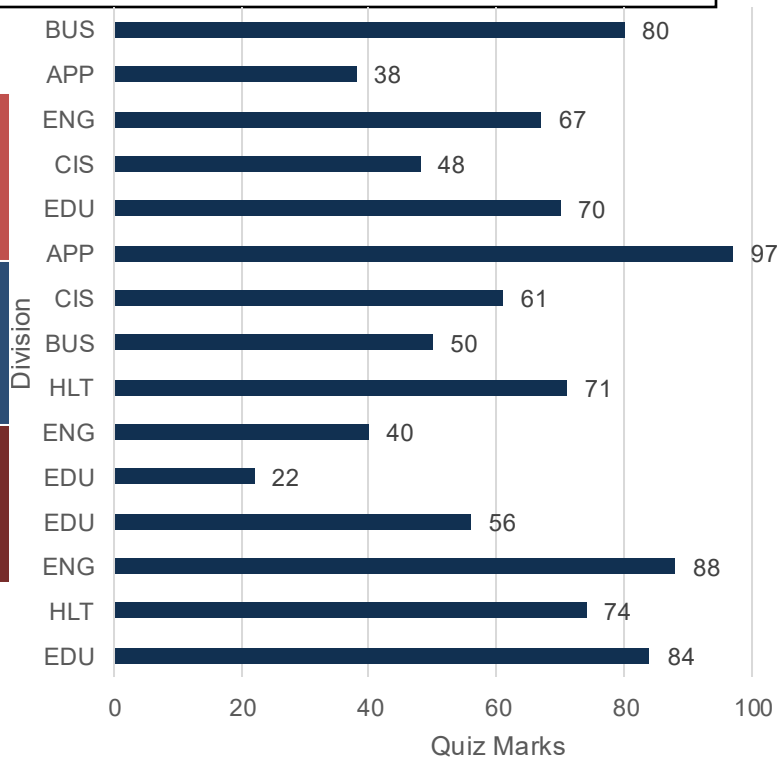
What is Data Visualization?



Histogram



Tree Map



Column Chart

Are you now able to assess which division performed well in Quiz, or Project, or Final marks?

Types of Charts/Graphs

Chart Type	Python Library
Scatter plot	Matplotlib, Seaborn, Altair
Line chart	Matplotlib, Seaborn, Altair
Bar chart	Matplotlib, Seaborn, Altair
Histogram	Matplotlib, Seaborn, Altair
Box plot	Matplotlib, Seaborn, Altair
Heatmap	Seaborn, Altair
Pie chart	Matplotlib, Seaborn
Area chart	Matplotlib, Seaborn, Altair
Violin plot	Seaborn, Altair

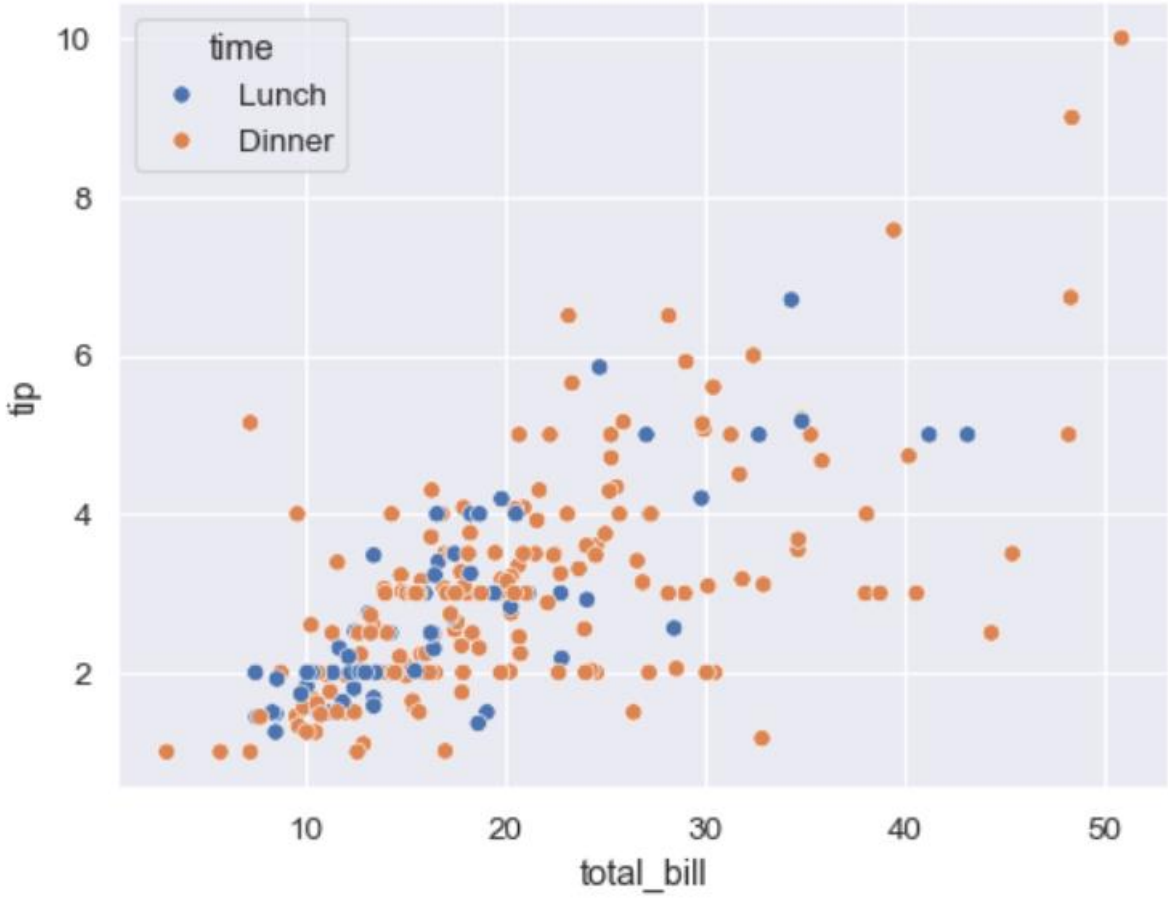
Types of Charts/Graphs

- Scatter Plot
- Box Plot
- Histogram
- Heat Map

Due to scope of this course, only limited graphs/charts will be analyzed

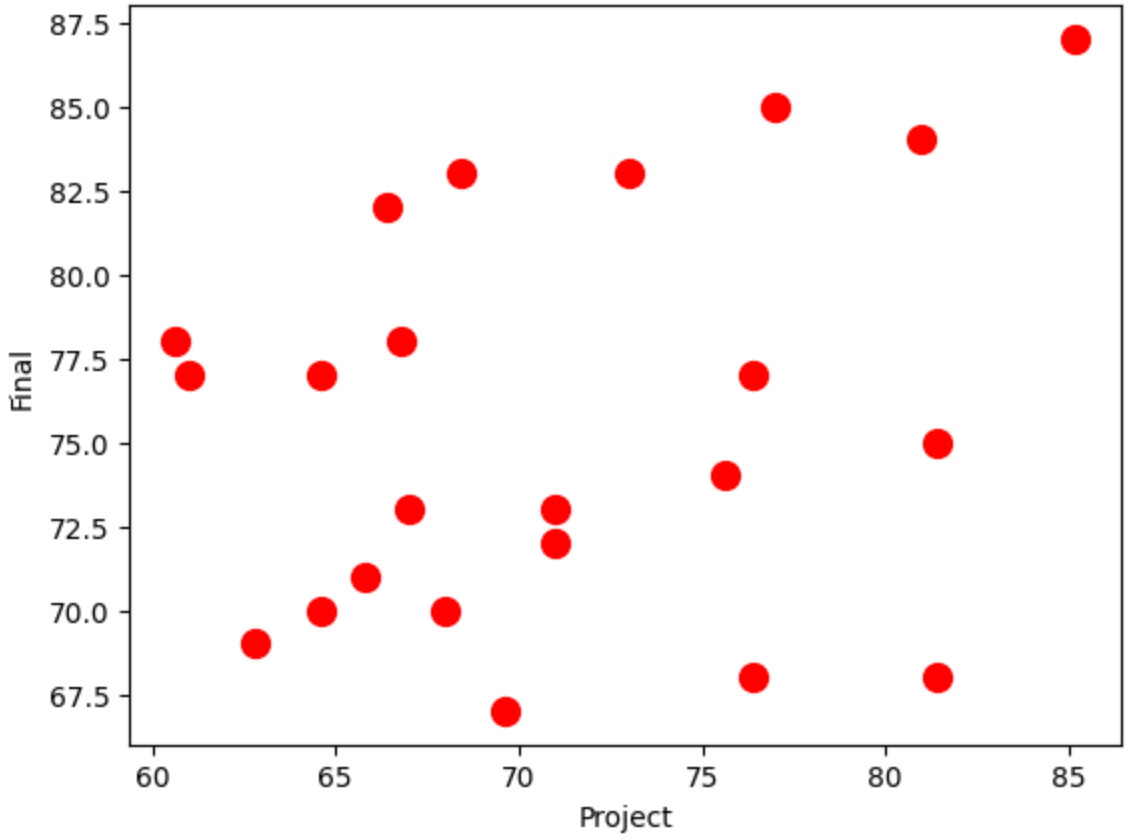
Types of Charts/Graphs - Scatter Plot

- A scatter plot is a graph that shows the relationship between two variables.
- The variables are plotted on the x-axis and y-axis, and the points are connected with lines.



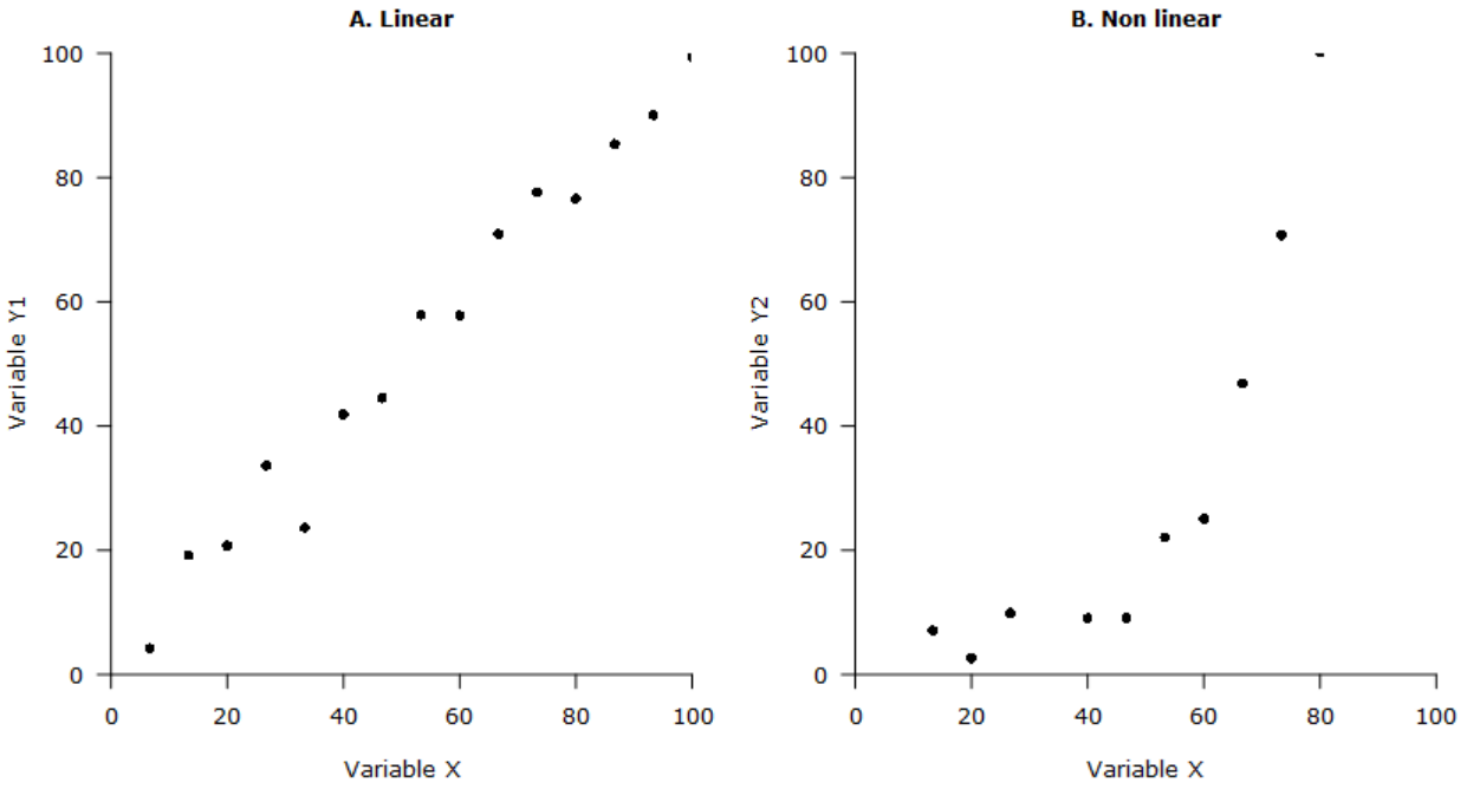
Types of Charts/Graphs - Scatter Plot

- The points are scattered, there is no correlation between the Project and Final marks variables.



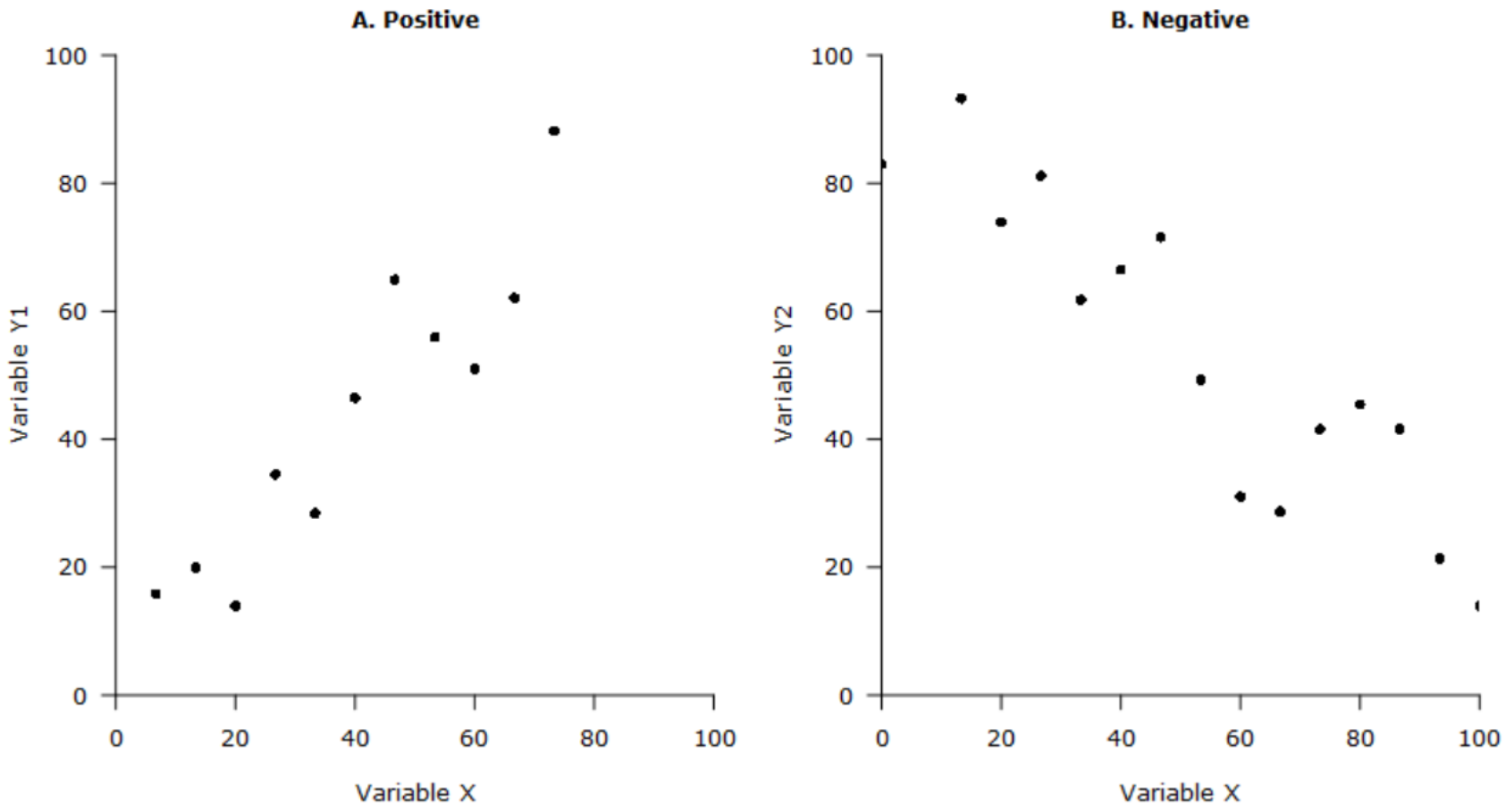
Types of Charts/Graphs - Scatter Plot

- Linear Vs Non-Linear



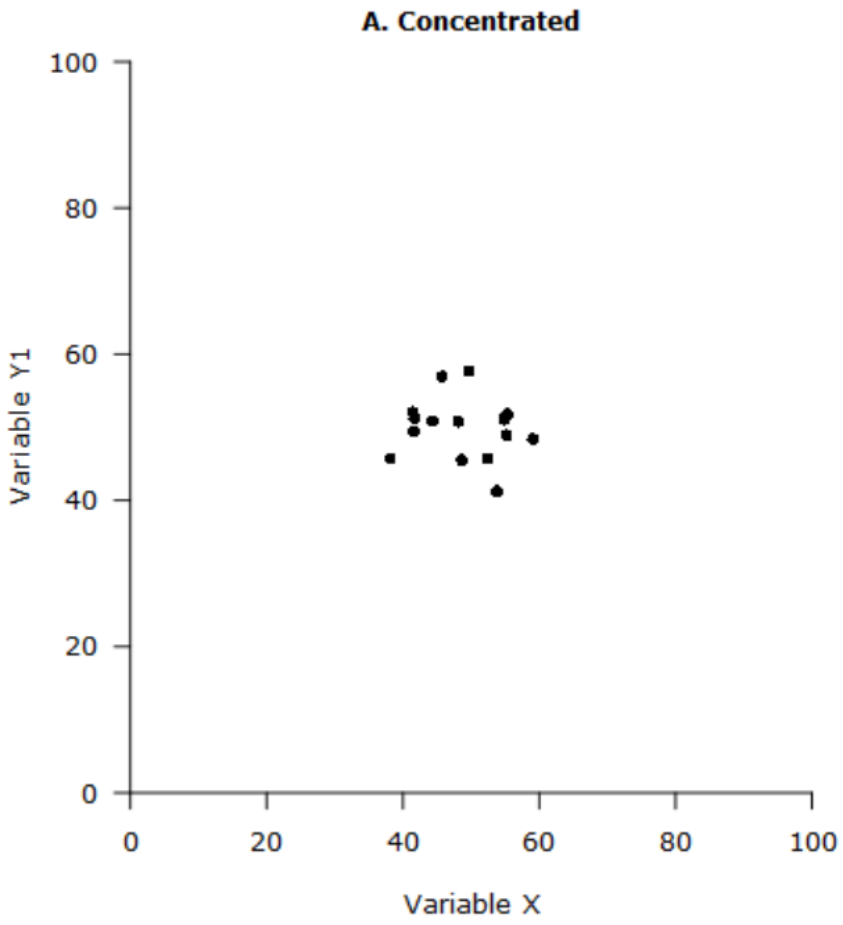
Types of Charts/Graphs - Scatter Plot

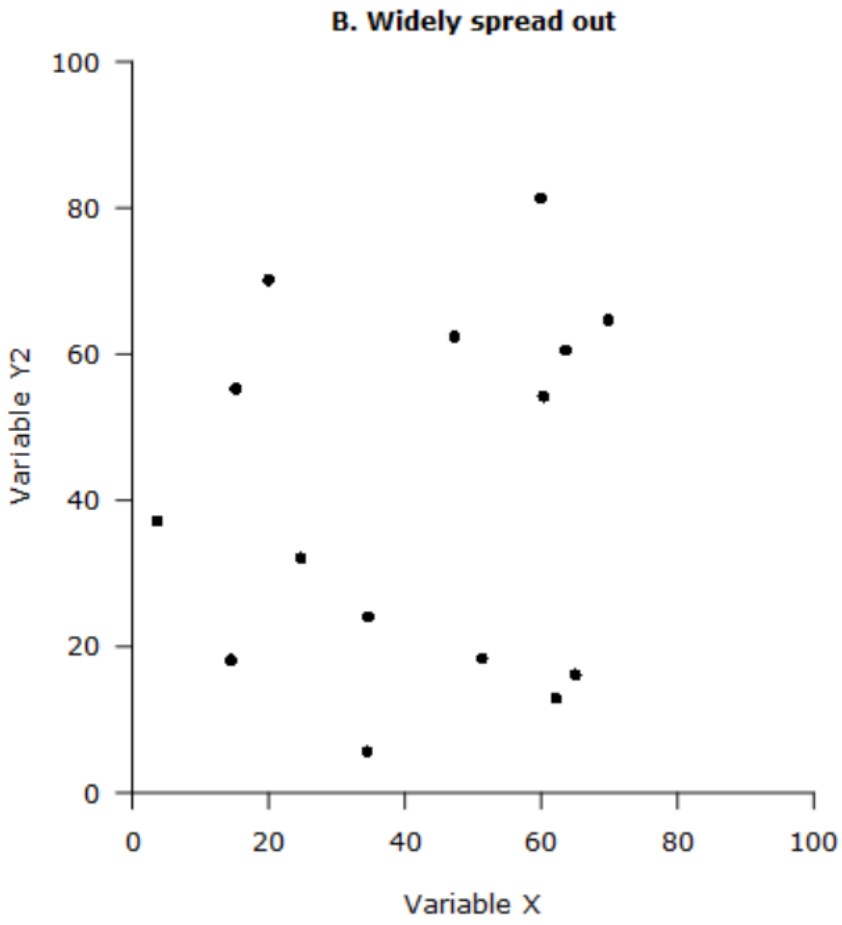
- Positive Vs Negative



Types of Charts/Graphs - Scatter Plot

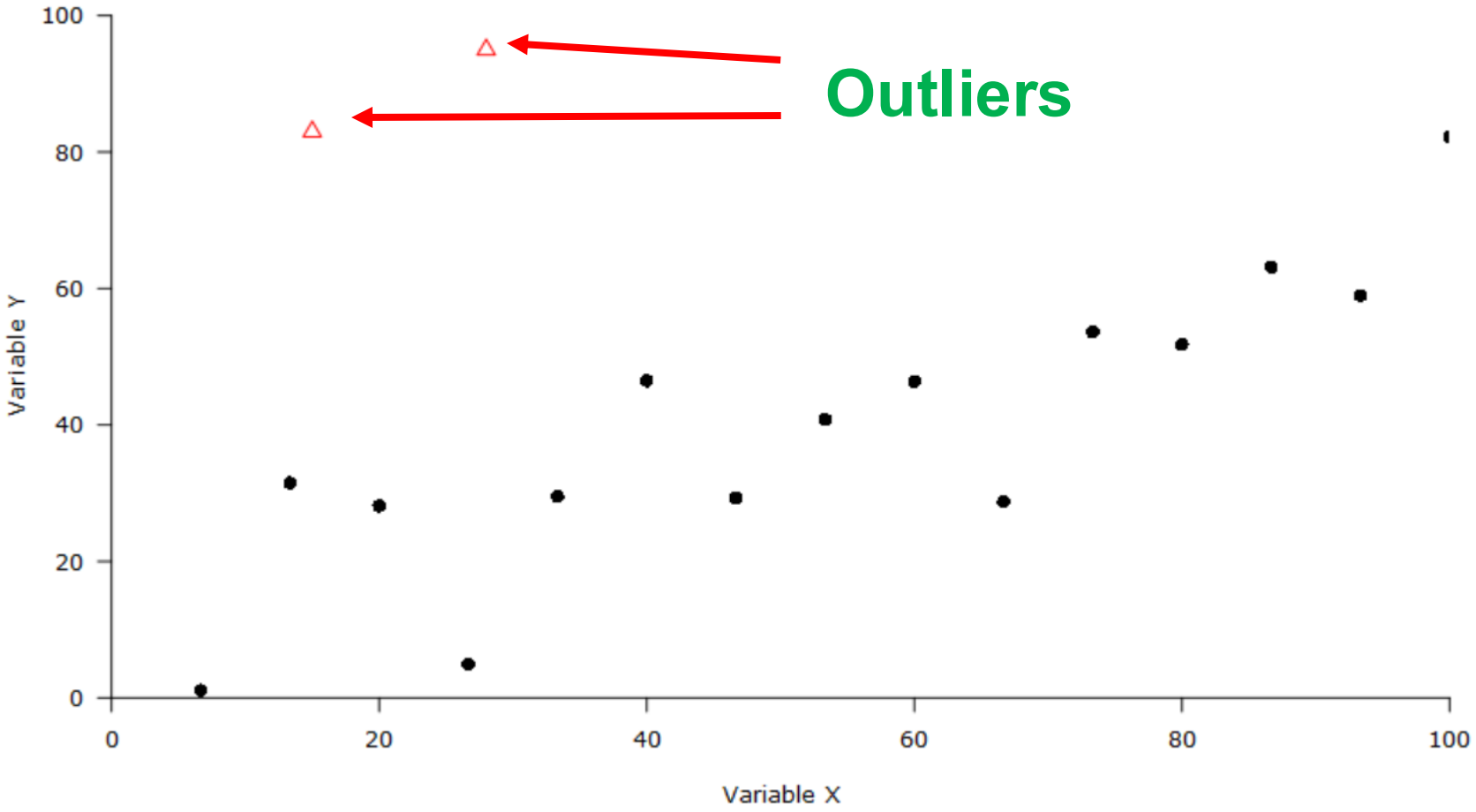
- Concentration Vs Spread
- More concentrated the dots, the stronger the relationship between the points





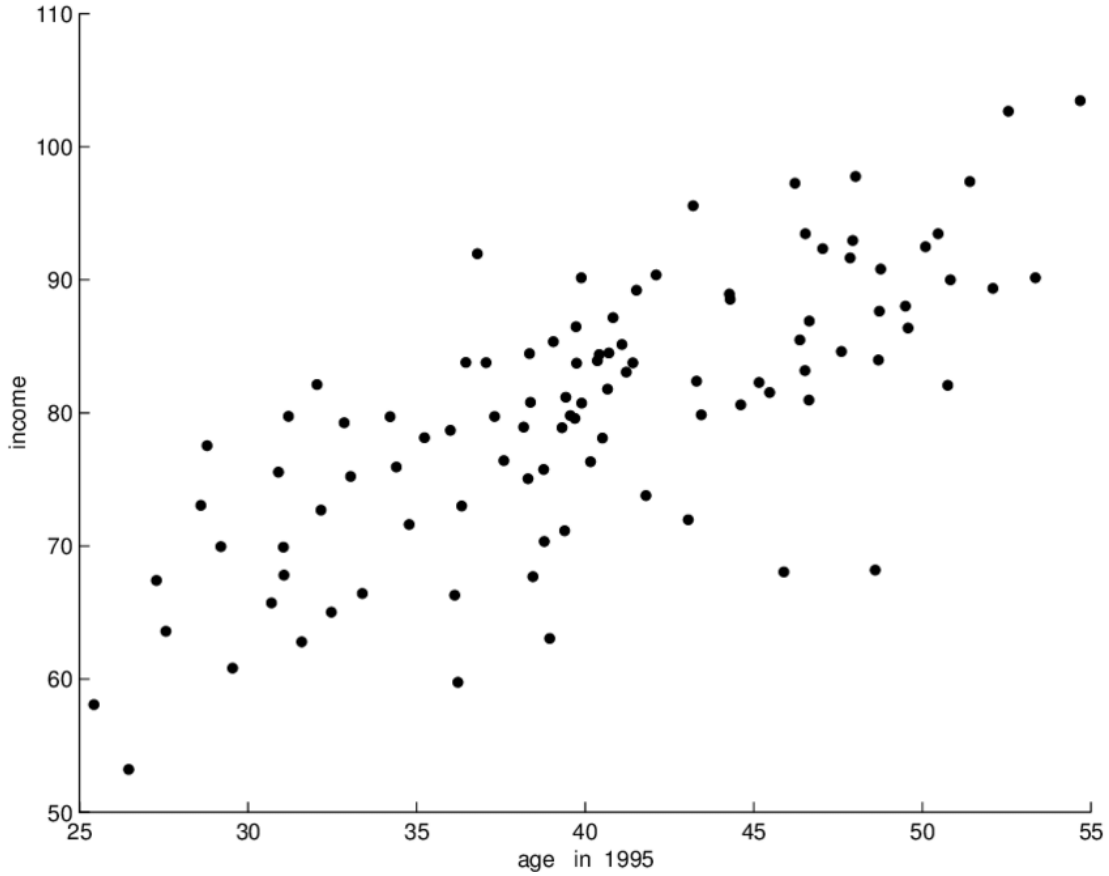
Types of Charts/Graphs - Scatter Plot

- Outlier detection and removal is a crucial data analysis step for a machine learning model, as outliers can significantly impact the accuracy of a model if they are not handled properly



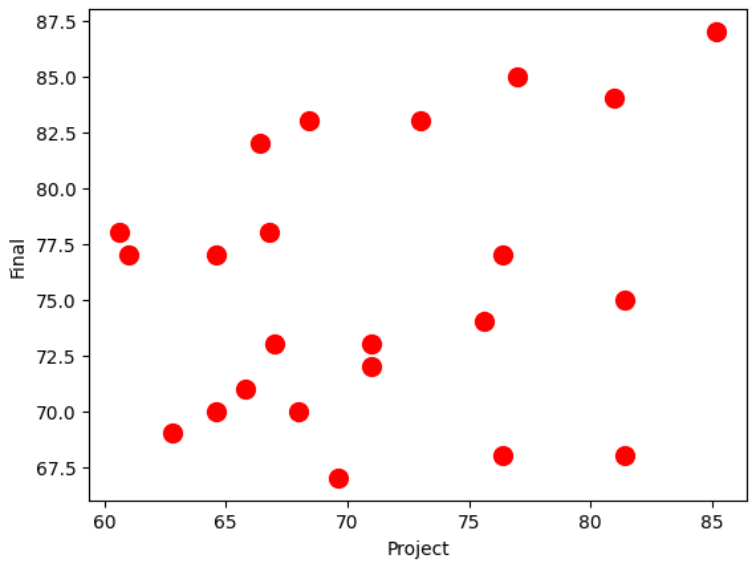
Types of Charts/Graphs – Scatter Plot - Quiz

- Based on the scatter plot, find
  - Is age and income correlated?
  - Is Linear or Non-linear?
  - Is positive or negative?
  - Is Concentrated or Spread-out?
  - Any outlier?



Types of Charts/Graphs - Scatter Plot-Python

```python
import pandas as pd
import matplotlib.pyplot as plt

# load the data into a pandas dataframe
df = pd.read_csv('Marks.csv')

# create a scatter plot for Project Vs Final marks
df.plot.scatter(x = 'Project', y = 'Final', s = 100, c = 'red');
```



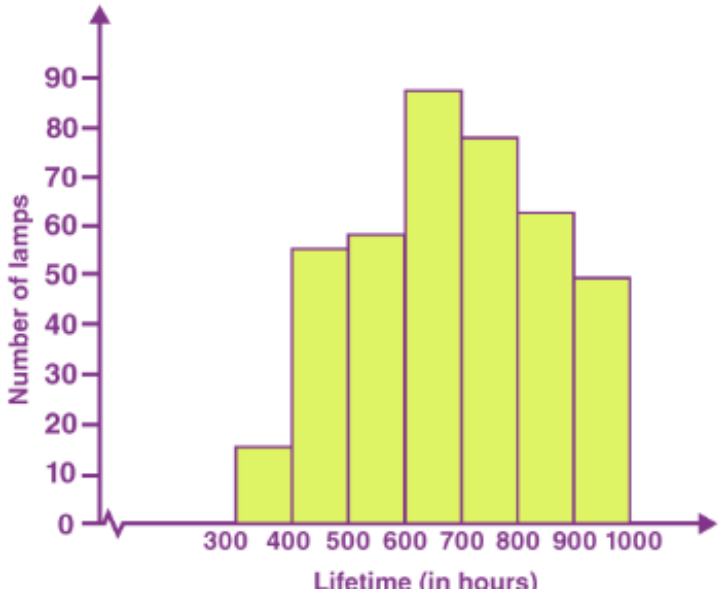
Size of the plot(circle)

Color of the plot

Write a python program to draw scatter plot between project and final marks. Critically analyse (Interpret) the results.

Types of Charts/Graphs - Histogram

- A histogram is a display of statistical information that uses rectangles(bins) to show the frequency of data items in successive numerical intervals of equal size.
- In the most common form of histogram,
  - the variable is plotted along the horizontal axis and
  - the frequency is plotted along the vertical axis

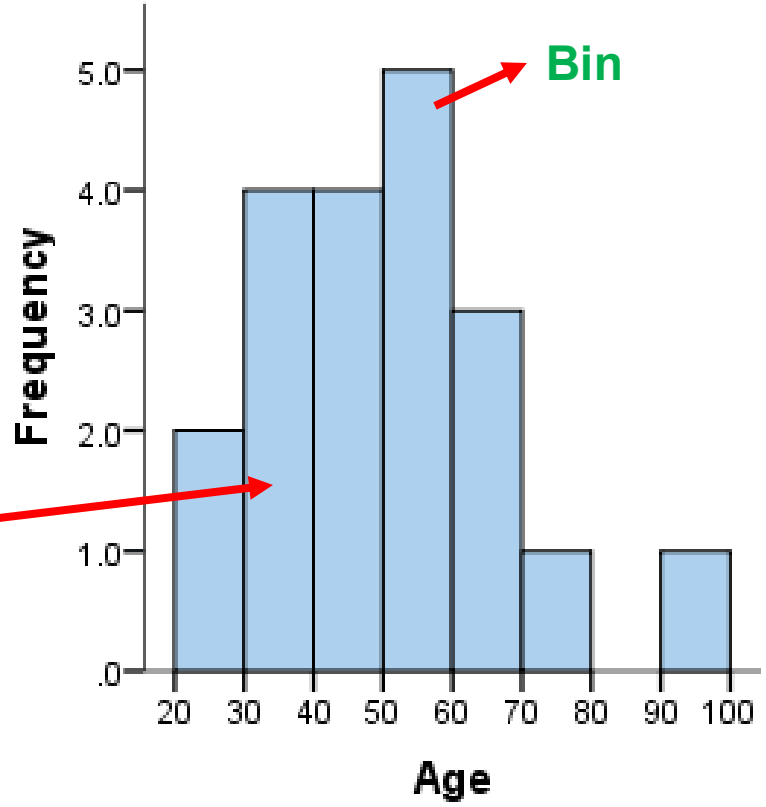


Types of Charts/Graphs - Histogram

- An example of a histogram, and the raw data it was constructed from, is shown below:

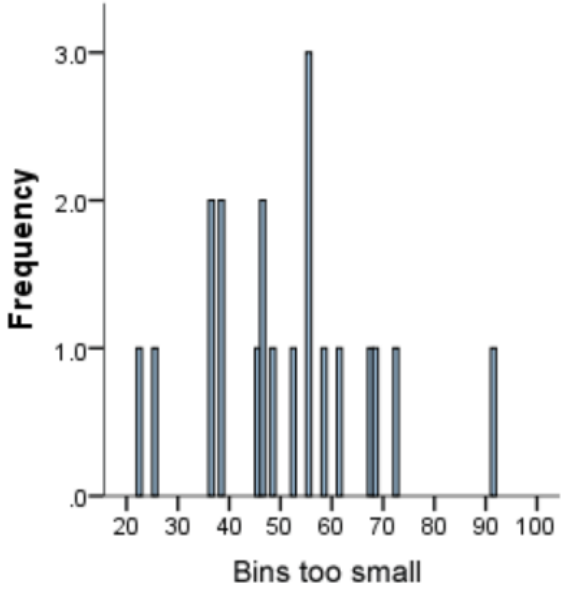
36	25	38	46	55	68	72	55	36	38
67	45	22	48	91	46	52	61	58	55

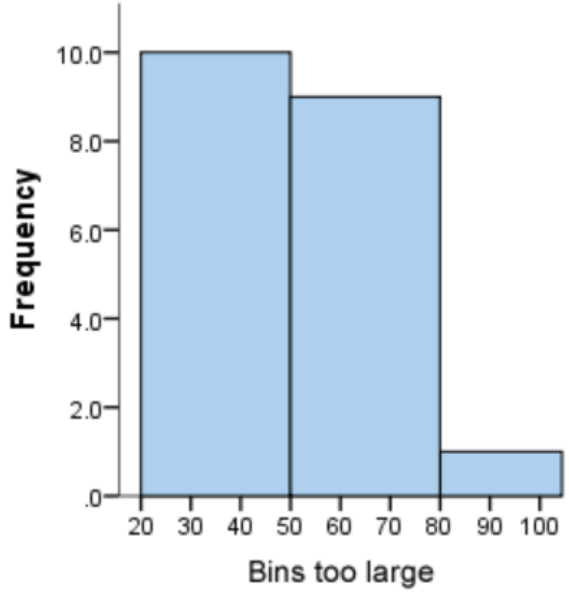
Bin	Frequency	Scores Included in Bin
20-30	2	25,22
30-40	4	36,38,36,38
40-50	4	46,45,48,46
50-60	5	55,55,52,58,55
60-70	3	68,67,61
70-80	1	72
80-90	0	-
90-100	1	91



Types of Charts/Graphs - Histogram

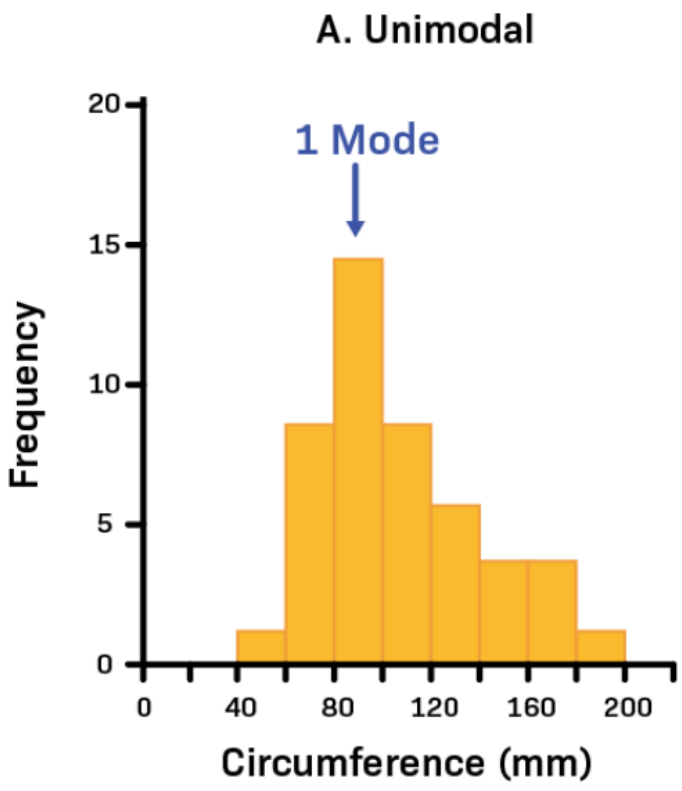
- The width of a bin is subjective, but guidelines suggest that bins should not be too small or too large to reveal the underlying pattern or trend in the data or to display too much individual data.

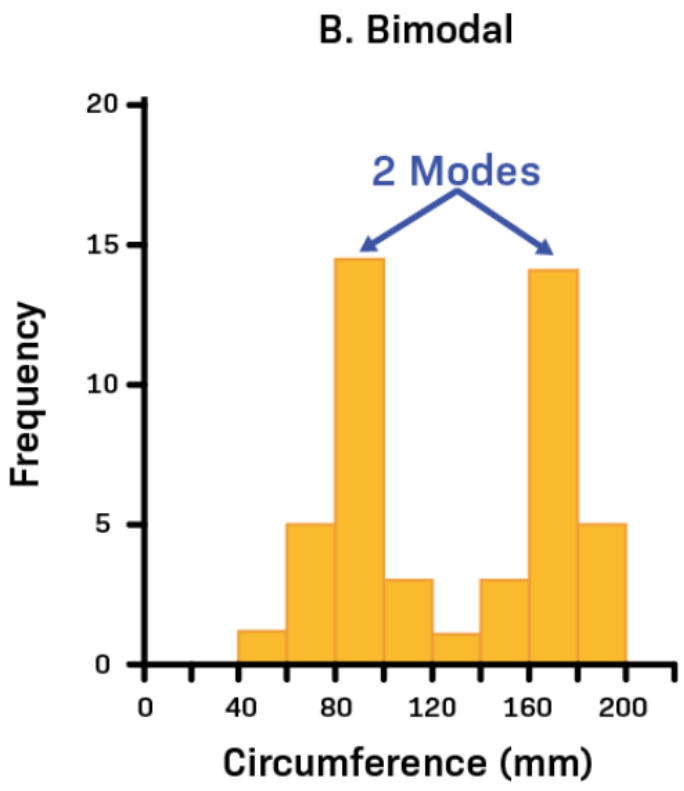




Types of Charts/Graphs - Histogram

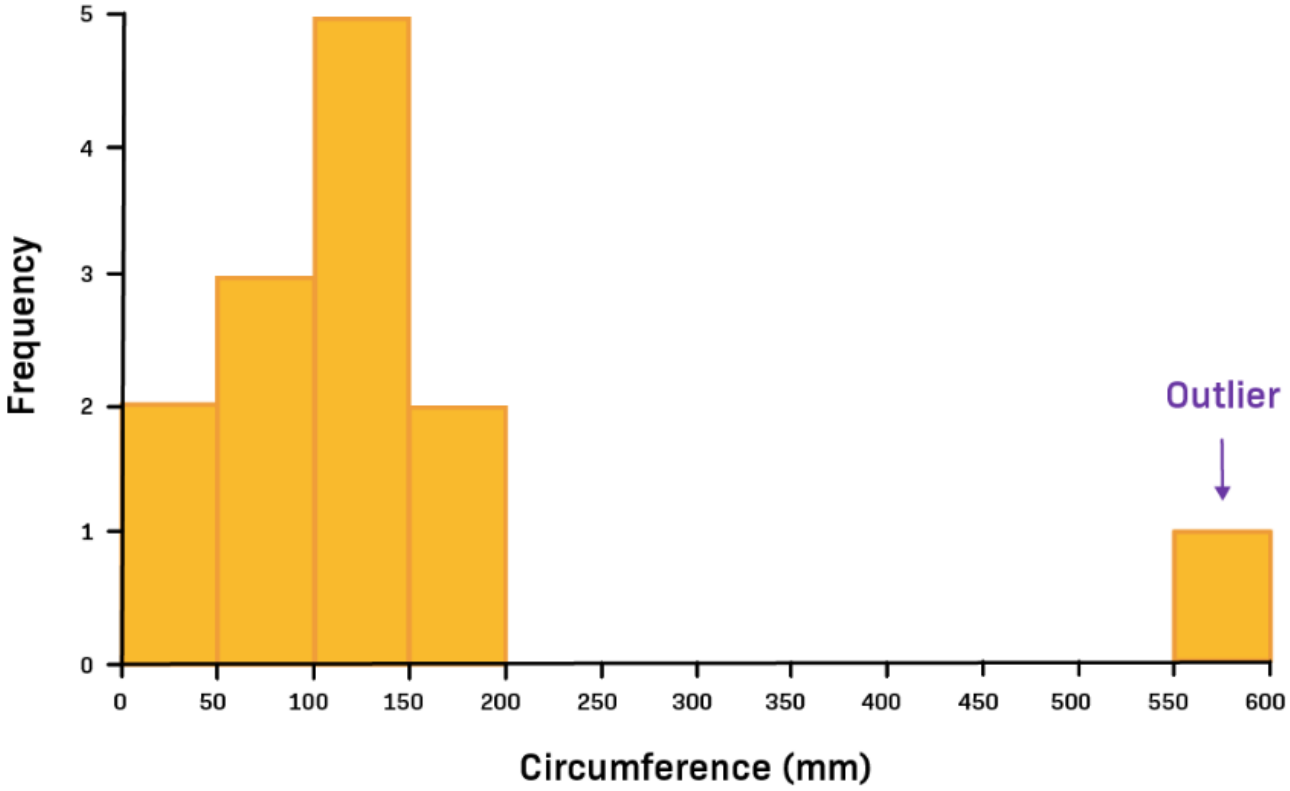
- The number of modes in a data set can be estimated by the number of prominent peaks present in a histogram.





Types of Charts/Graphs - Histogram

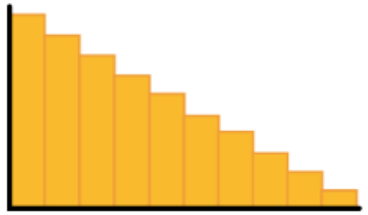
- Outlier are the data points which are far away from other data points
- If outlier exists, then it should be removed or mean of the data should be placed from the dataset.
- Otherwise, data analysis will not be effective



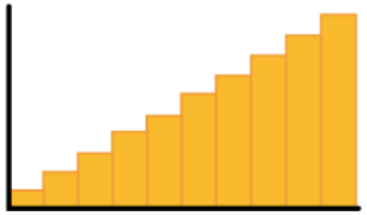
Types of Charts/Graphs - Histogram

Symmetric (normal) vs skewed and uniform distriutions

Normal distribution
(unimodal, symmetric,
the "bell curve")



Right-skewed distribution
(Positively-skewed)



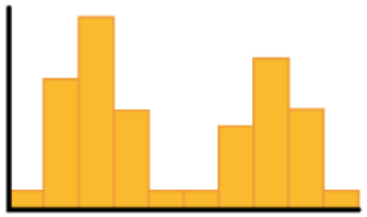
Left-skewed distribution
(Negatively-skewed)

Uniform distribution
(equal spread,
no peaks)

Unimodal vs bimodal distributions

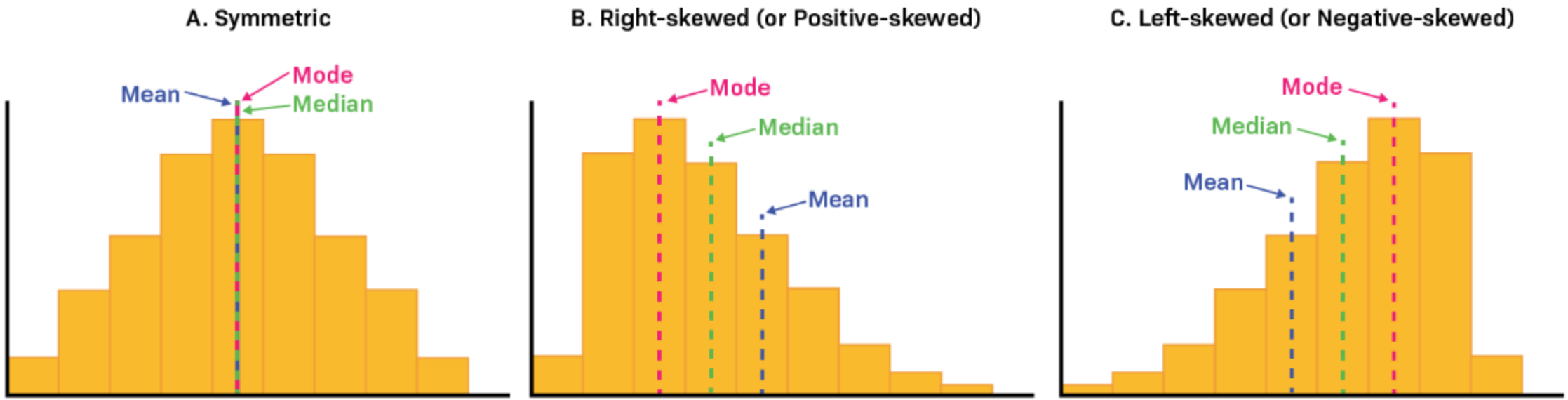
Normal distribution
(unimodal, symmetric,
the "bell curve")

Symmetric bimodal distribution
(two modes)



Non-symmetric bimodal distribution
(two modes)

Types of Charts/Graphs - Histogram



Types of Charts/Graphs - Histogram – Case Study

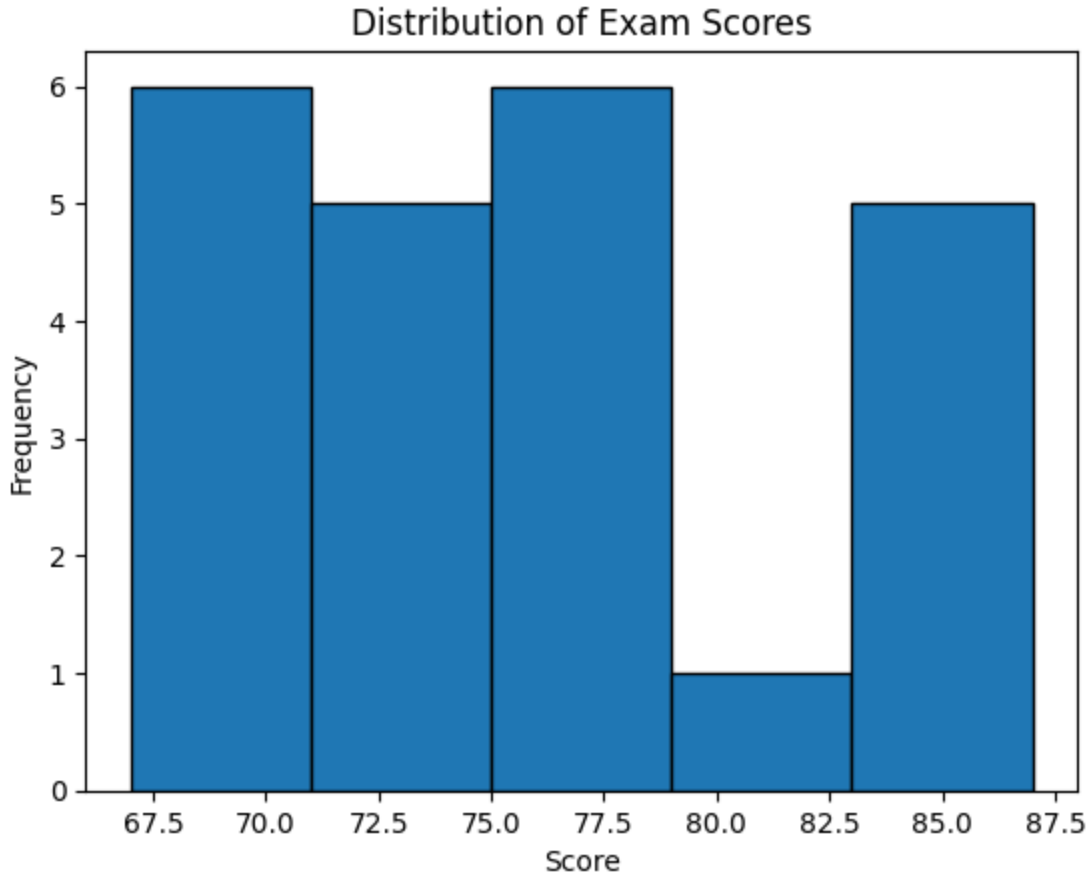
- A 1000 student exam dataset is analyzed using a histogram, dividing it into 5 equally sized bins.
- The bins=5 parameter helps visualize the distribution and identify trends or outliers.
- Matplot library is used for drawing histogram

Perform histogram for FWA

```
import pandas as pd
import matplotlib.pyplot as plt

# load the data into a pandas dataframe
df = pd.read_csv('Marks.csv')

# create a histogram
plt.hist(df['Final'], bins=5, edgecolor='black')

# add labels and title
plt.xlabel('FWA')
plt.ylabel('Frequency')
plt.title('Distribution of Exam Scores')

# display the plot
plt.show()
```

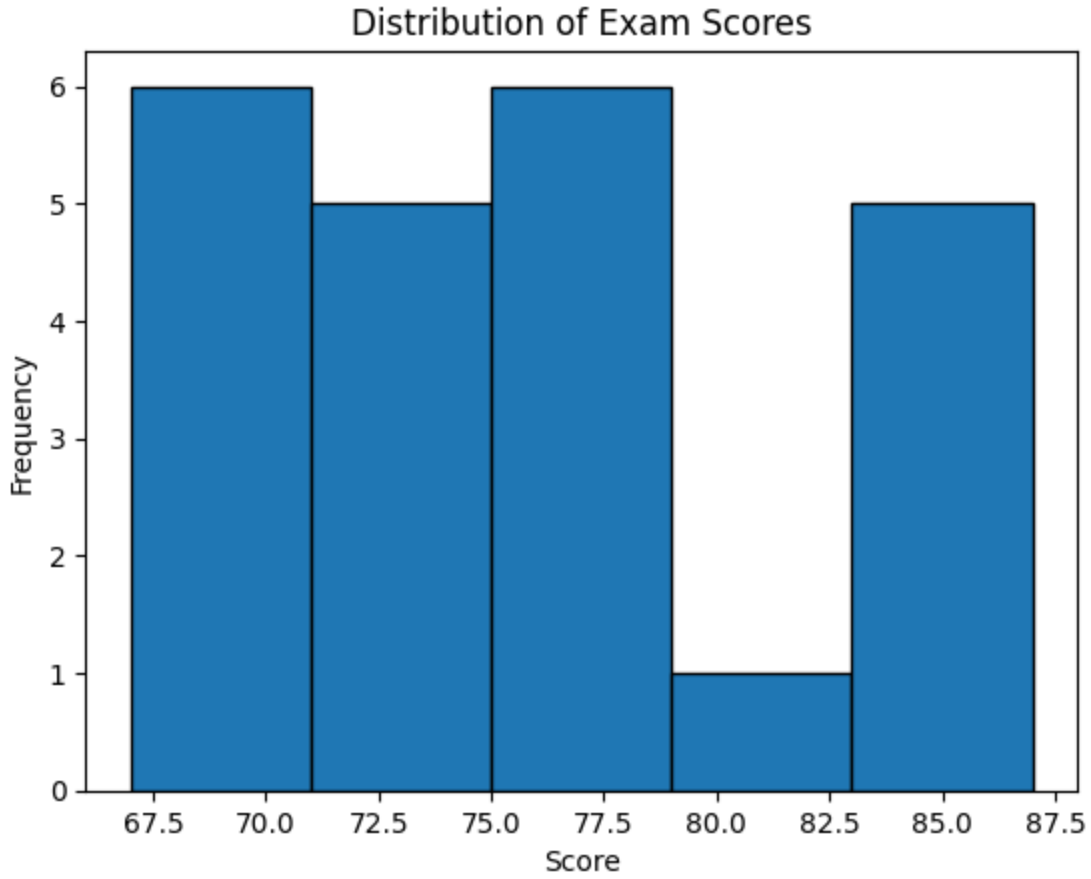
Types of Charts/Graphs - Histogram– Quiz

- Interpret using the given histogram:
  - How many students score between 83 to 87?
  - How many students score between 67 to 71?
  - What is the maximum mark?
  - What is the minimum mark?
  - How many modes are there?
  - Are marks positively or negatively skewed or uniformly distributed?



Types of Charts/Graphs - Histogram– Case Study

- Around 5 students score between 83 to 87
- Only one student score between 79 to 83
- Around 6 students score between 75 to 79
- Around 5 students score between 71 to 75
- Around 6 students score between 67 to 71
- Right – Skewed
- Min – 67 and Max – 87



Types of Charts/Graphs - Box Plot

- A box plot shows the distribution of data through:
  - Minimum, maximum, and median values.
  - Lower and upper quartiles.
  - Allows comparison between datasets.
- From this, the range, interquartile range and skewness of the data can be observed.
- Box plots provide a visual summary of data distribution and can highlight outliers.

Types of Charts/Graphs - Box Plot

- Consider the **sorted** salary data with 20 rows.
- First Quartile(Q1)- 25% - 5th row
- Second Quartile (Q2) - 50% - Median - 10th row
- Third Quartile(Q3) – 75% - 15th row)

	Employee ID	Salary (in thousands of dollars)
1	10	22
2	5	23
3	12	23
4	20	23
5	1	24
6	7	24
7	16	24
8	3	25
9	13	25
10	2	26
11	11	26
12	17	26
13	6	27
14	14	27
15	18	27
16	4	28
17	15	28
18	8	29
19	19	29
20	9	30

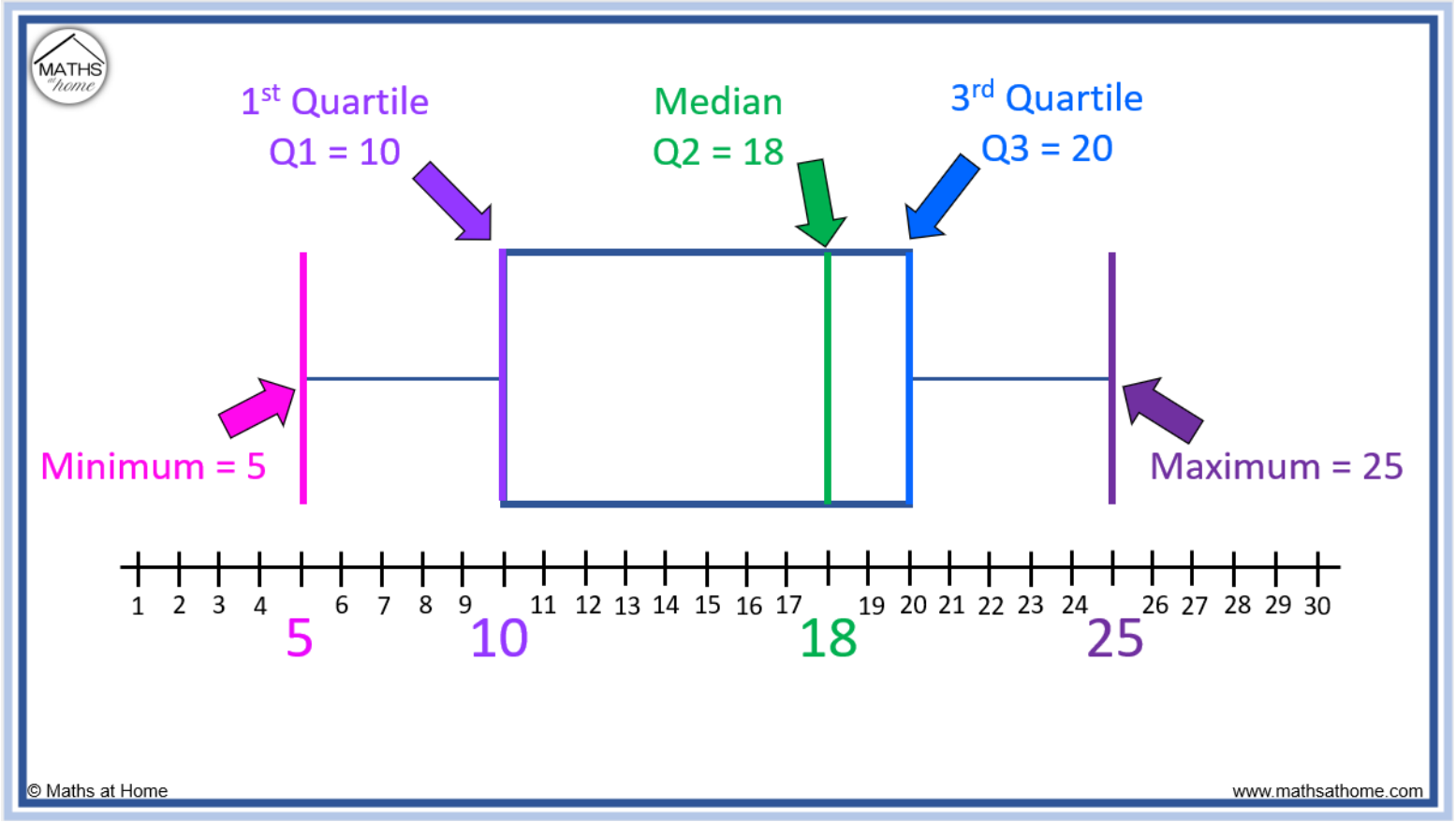
Types of Charts/Graphs - Box Plot

- Q1 - 25% of employees salaries are below $ 24,000
- Q2 - 50% of employees salaries are below $ 26,000
- Q3 – 75% of employees salaries are below $ 27,000

	Employee ID	Salary (in thousands of dollars)
1	10	22
2	5	23
3	12	23
4	20	23
5	1	24
6	7	24
7	16	24
8	3	25
9	13	25
10	2	26
11	11	26
12	17	26
13	6	27
14	14	27
15	18	27
16	4	28
17	15	28
18	8	29
19	19	29
20	9	30

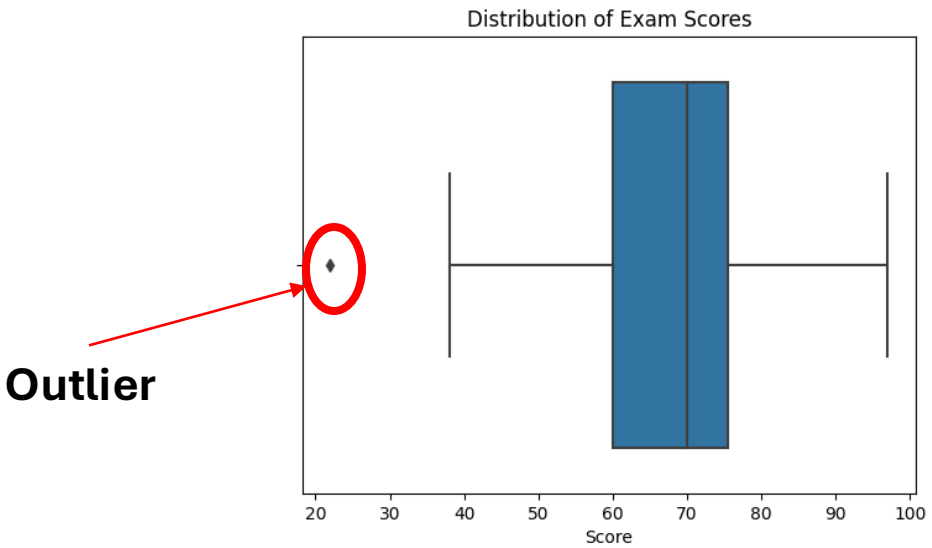
Types of Charts/Graphs - Box Plot

- **To read a box plot:**
- Read the minimum value in line with the first line.
- Read the maximum value in line with the last line.
- Read the lower quartile which is in line with the start of the box.
- Read the upper quartile which is in line with the end of the box.
- Read the median which is in line with the line inside the box.



Types of Charts/Graphs - Box Plot- Python

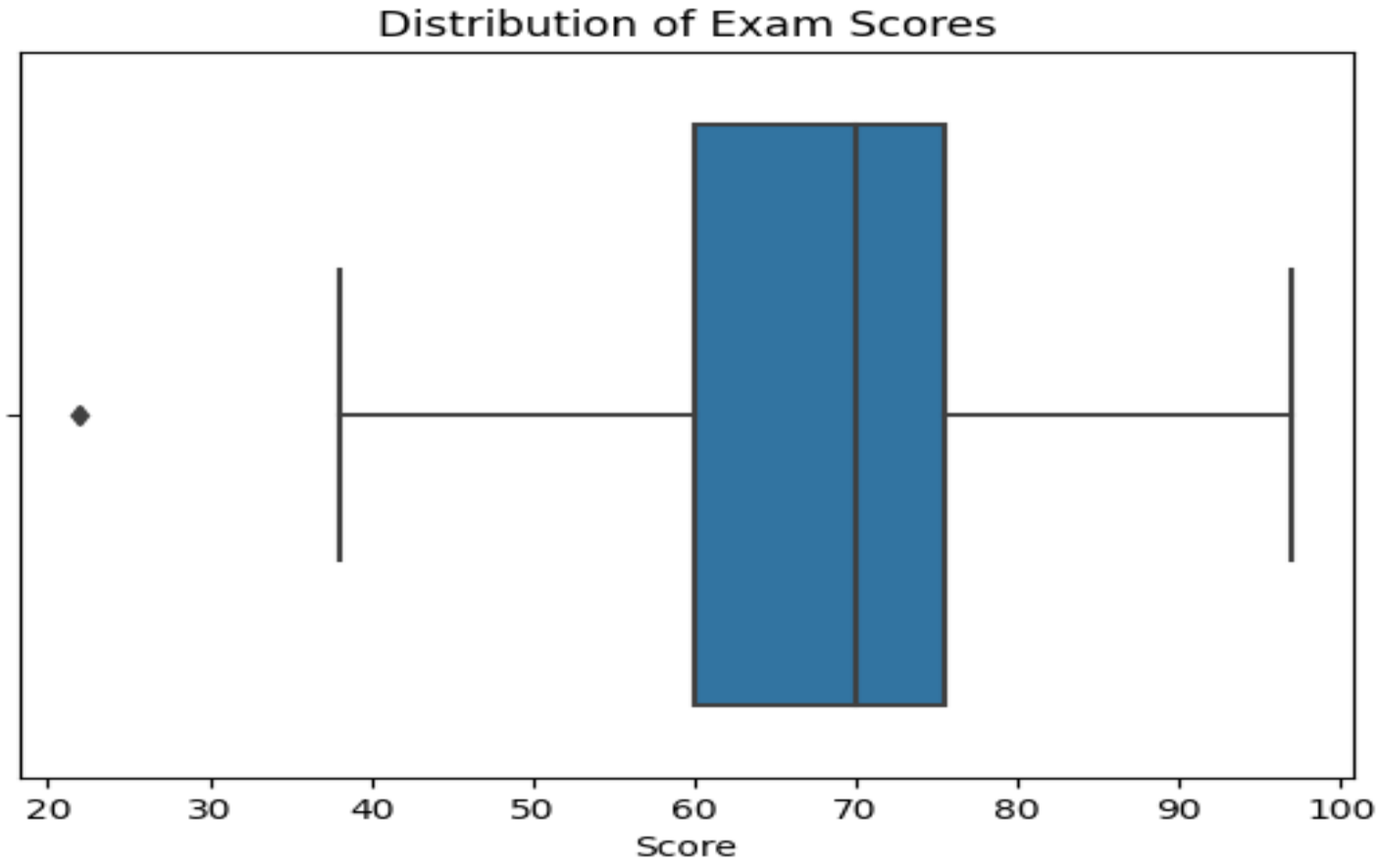
- Use Seaborn library to plot Box Plot
- The x=df['Quiz'] parameter specifies that we want to create a box plot of the Quiz column in the data frame.




```python
import pandas as pd
import seaborn as sns

# load the data into a pandas dataframe
df = pd.read_csv('Marks.csv')

# create a box plot
sns.boxplot(x=df['Quiz'])

# add labels and title
plt.xlabel('Score')
plt.title('Distribution of Exam Scores')

# display the plot
plt.show()
```

Types of Charts/Graphs - Box Plot

- **Read a box plot and find:**
- Minimum mark for the Quiz is –
- Maximum mark for the Quiz is -
- 25% of students score less than -
- 50% of students score less than -
- 75% of students score less than –



Types of Charts/Graphs - Heat Map

- A heat map represents data values with different colors:
  - Darker colors typically indicate higher values.
  - Used to identify patterns, outliers, and trends in data.
- Heat maps are effective for showing data intensity across different areas or time periods.
- Seaborn library is used to draw Heat map in python

Types of Charts/Graphs - Heat Map

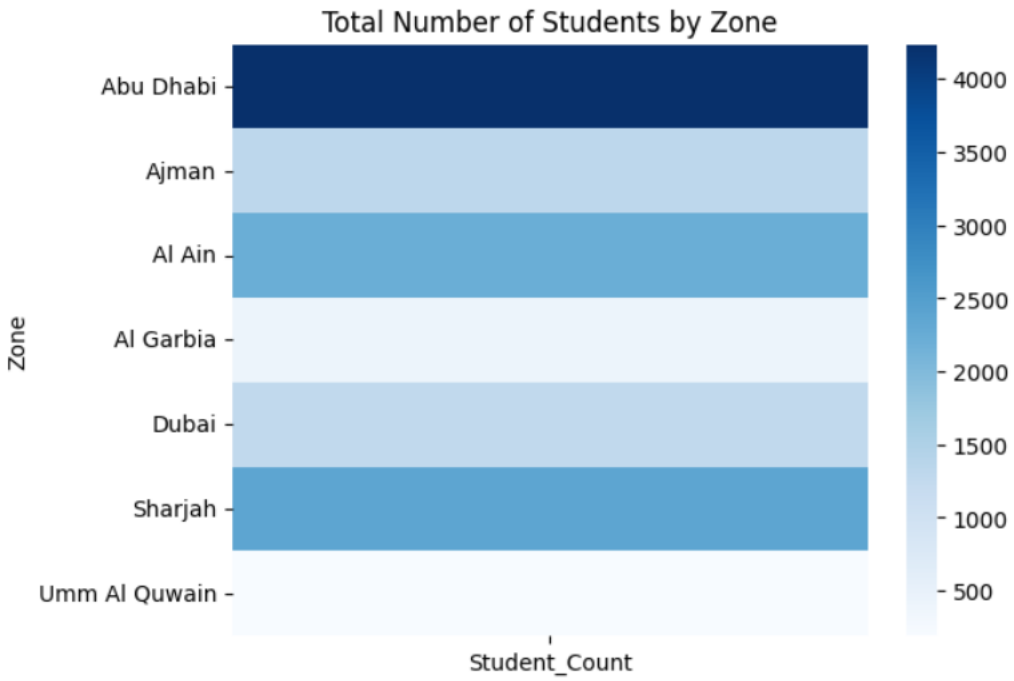
- Plot a heat map for number of students by zone

Year	Category_EN	Zone_EN	Stage_EN	Gender_E	Student_Count
2015/2016	Government Education	Abu Dhabi	Kindergarten	Male	2087
2015/2016	Government Education	Abu Dhabi	Kindergarten	Female	2150
2015/2016	Government Education	Al Ain	Kindergarten	Male	1724
2015/2016	Government Education	Al Ain	Kindergarten	Female	500
2015/2016	Government Education	Al Garbia	Kindergarten	Male	200
2015/2016	Government Education	Al Garbia	Kindergarten	Female	225
2015/2016	Government Education	Dubai	Kindergarten	Male	620
2015/2016	Government Education	Dubai	Kindergarten	Female	652
2015/2016	Government Education	Sharjah	Kindergarten	Male	1152
2015/2016	Government Education	Sharjah	Kindergarten	Female	1223
2015/2016	Government Education	Ajman	Kindergarten	Male	900
2015/2016	Government Education	Ajman	Kindergarten	Female	429
2015/2016	Government Education	Umm Al Quwain	Kindergarten	Male	193

Types of Charts/Graphs - Heat Map– Python

- Plot a heat map for number of students by zone

```python
# libraries
import pandas as pd
import seaborn as sns
import numpy as np
import matplotlib.pyplot as plt

# create data
df = pd.read_csv('Students.csv')

# plot
pivot_table = pd.pivot_table(df, index='Zone_EN', values='Student_Count', aggfunc='sum')
sns.heatmap(pivot_table, cmap='Blues')

plt.ylabel('Zone')
plt.title('Total Number of Students by Zone')

plt.show()
```



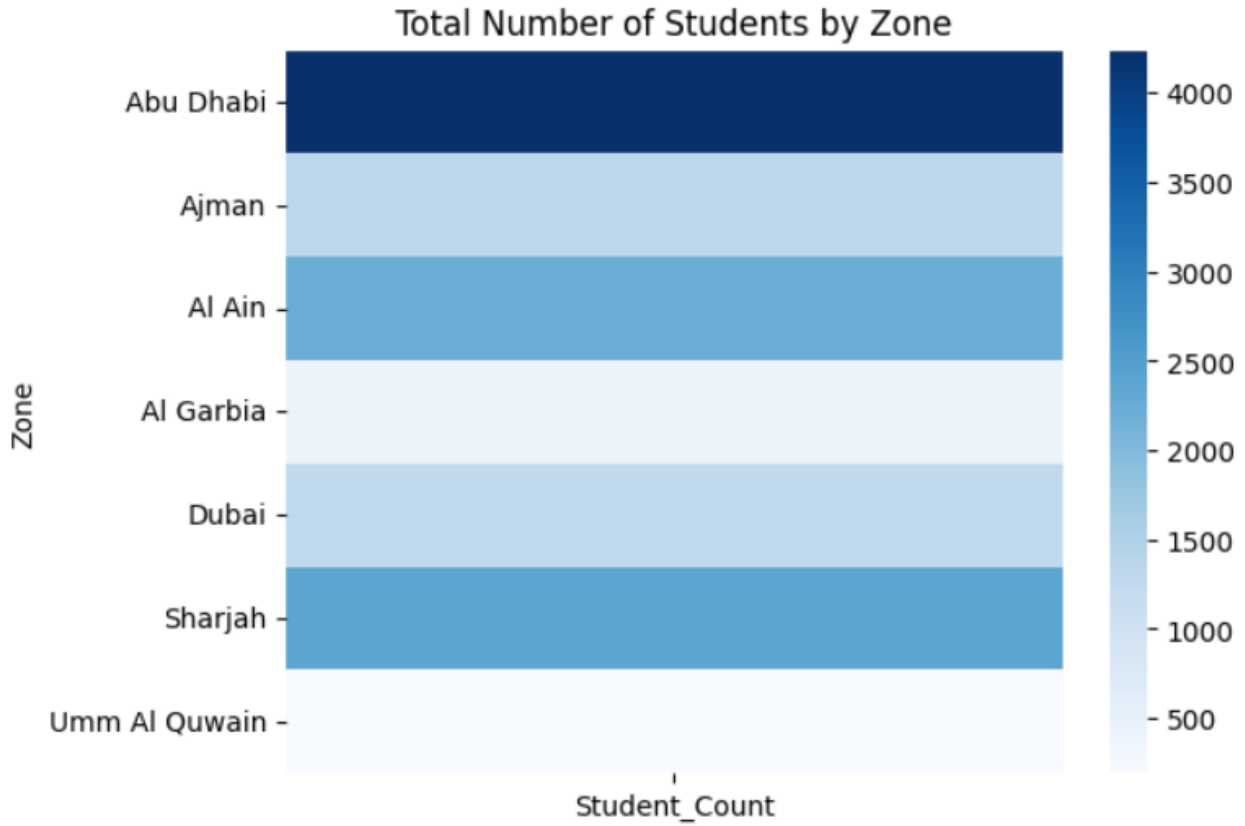
Types of Charts/Graphs - Heat Map– Python

Annotate the graph based on:

Specific Zones:

- **Highest Student Count:** Abu Dhabi (>4,000 students).
- **Lowest Student Count:** Umm Al Quwain (<500 students).
- **Intermediate Regions:** Al Ain and Sharjah have relatively high student counts, while Dubai and Ajman fall in the mid-range.
- **Least Populated Zones:** Al Garbia and Umm Al Quwain.
- **Dark Color:** Indicates a high student count (e.g., Abu Dhabi with over 4,000 students).
- **Light Color:** Indicates a low student count (e.g., Umm Al Quwain with fewer than 500 students).

- Plot a heat map for number of students by zone



Types of Charts/Graphs - Heat Map

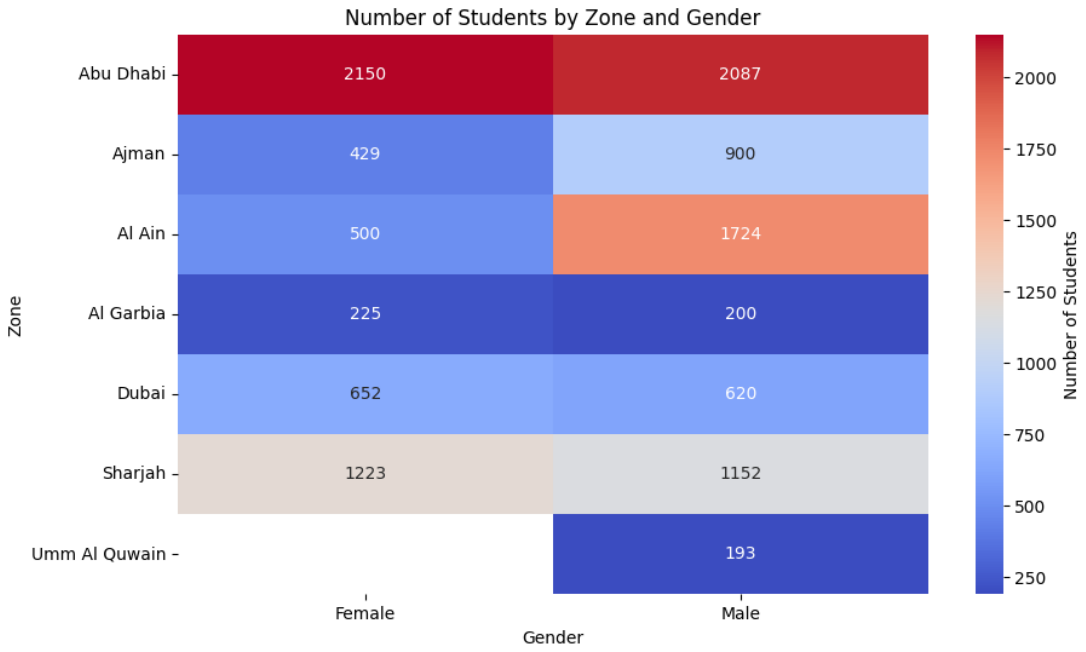
- Plot a heat map for number of students by zone and gender

Year	Category_EN	Zone_EN	Stage_EN	Gender_E	Student_Count
2015/2016	Government Education	Abu Dhabi	Kindergarten	Male	2087
2015/2016	Government Education	Abu Dhabi	Kindergarten	Female	2150
2015/2016	Government Education	Al Ain	Kindergarten	Male	1724
2015/2016	Government Education	Al Ain	Kindergarten	Female	500
2015/2016	Government Education	Al Garbia	Kindergarten	Male	200
2015/2016	Government Education	Al Garbia	Kindergarten	Female	225
2015/2016	Government Education	Dubai	Kindergarten	Male	620
2015/2016	Government Education	Dubai	Kindergarten	Female	652
2015/2016	Government Education	Sharjah	Kindergarten	Male	1152
2015/2016	Government Education	Sharjah	Kindergarten	Female	1223
2015/2016	Government Education	Ajman	Kindergarten	Male	900
2015/2016	Government Education	Ajman	Kindergarten	Female	429
2015/2016	Government Education	Umm Al Quwain	Kindergarten	Male	193

Types of Charts/Graphs - Heat Map– Python

- Plot a heat map for number of students by zone and gender

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load the CSV file
file_path = 'Students.csv'
students_data = pd.read_csv(file_path)

# Creating a pivot table to summarize data by Zone and Gender
heatmap_data = students_data.pivot_table(
    index='Zone_EN',
    columns='Gender_EN',
    values='Student_Count',
    aggfunc='sum'
)

# Plotting the heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(heatmap_data, annot=True, fmt=".0f", cmap="coolwarm", cbar_kws={'label': 'Number of Students'})
plt.title('Number of Students by Zone and Gender')
plt.ylabel('Zone')
plt.xlabel('Gender')
plt.show()
```



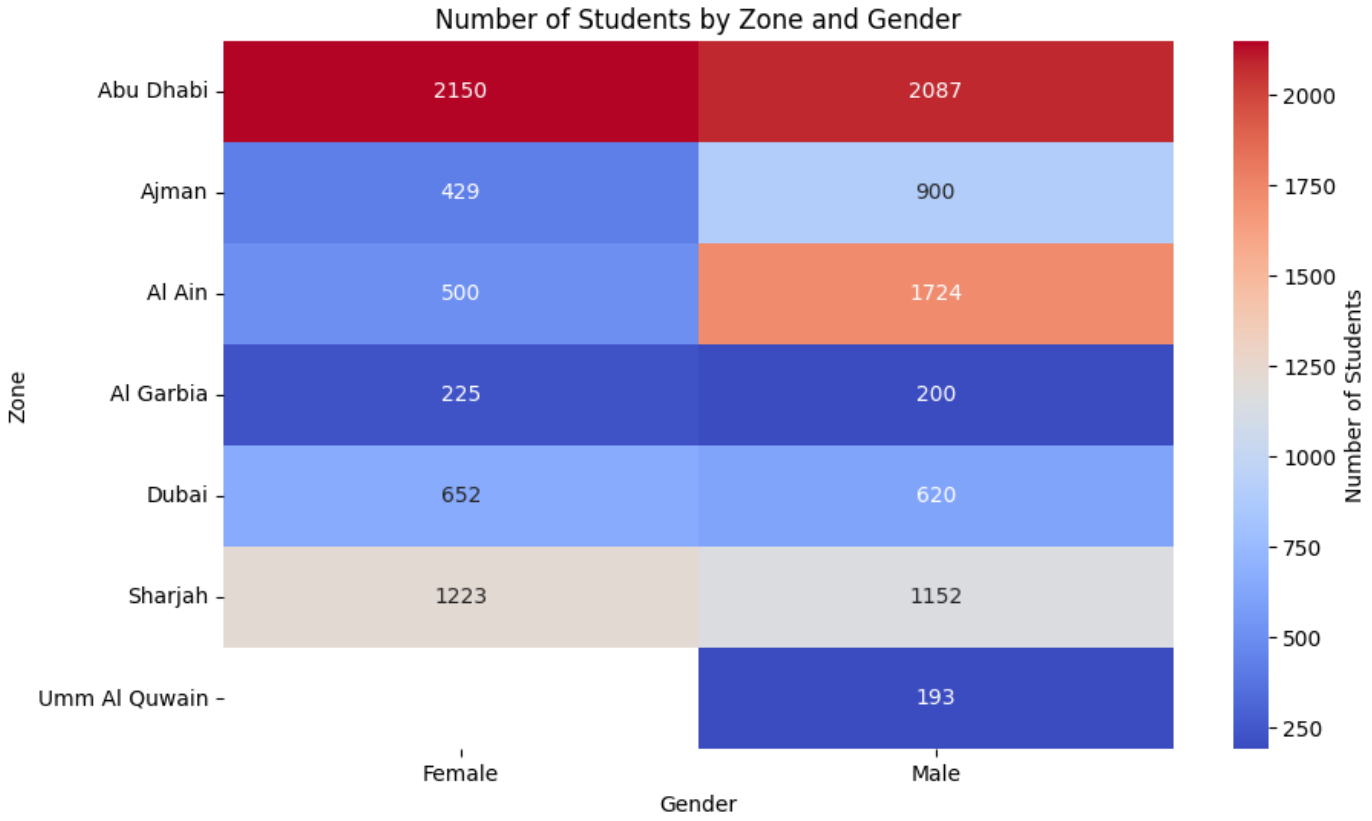
Types of Charts/Graphs - Heat Map– Python

Annotate the graph based on:

- Plot a heat map for number of students by zone and gender

Specific Zones:

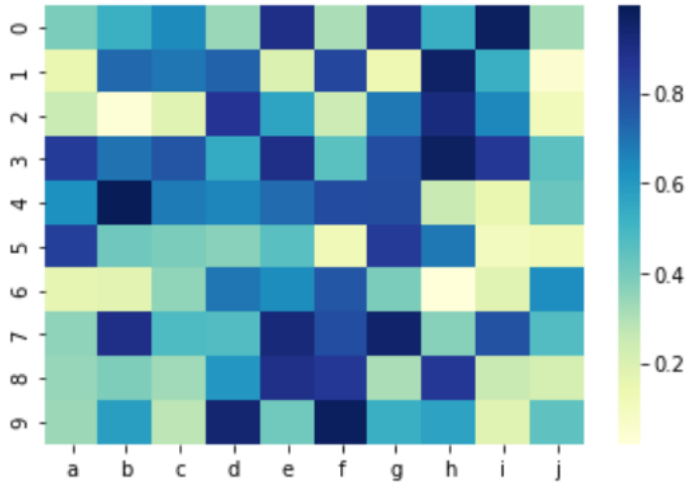
1. Abu Dhabi has almost equal numbers of female (2,150) and male (2,087) students, with a slight majority of female students.
2. Ajman has significantly more male (900) students compared to female (429) students.
3. Al Ain has a significantly higher number of male (1,724) students compared to female (500) students. It is not balanced.
4. Al Garbia has a slightly higher number of male (200) students compared to female (225) students.
5. Dubai has almost equal numbers of female (652) and male (620) students, with a slight majority of female students.
6. Sharjah has more female (1,223) students compared to male (1,152) students.
7. Umm Al Quwain has data only for female (193) students. There is no data for male students.

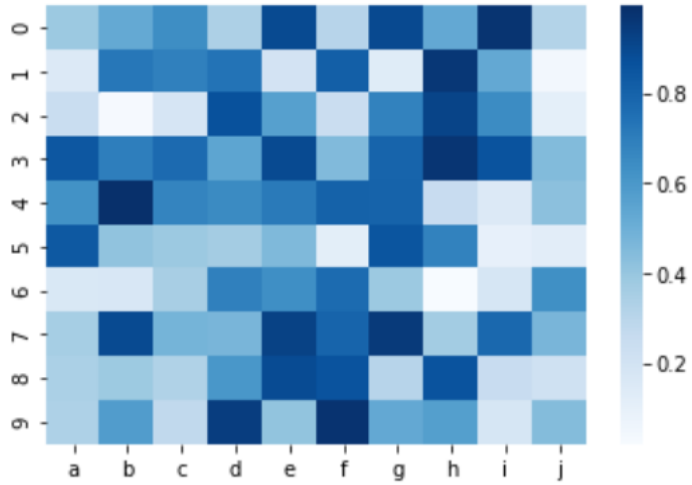


Types of Charts/Graphs - Heat Map – Python

Cmap parameter is used to shade different colours

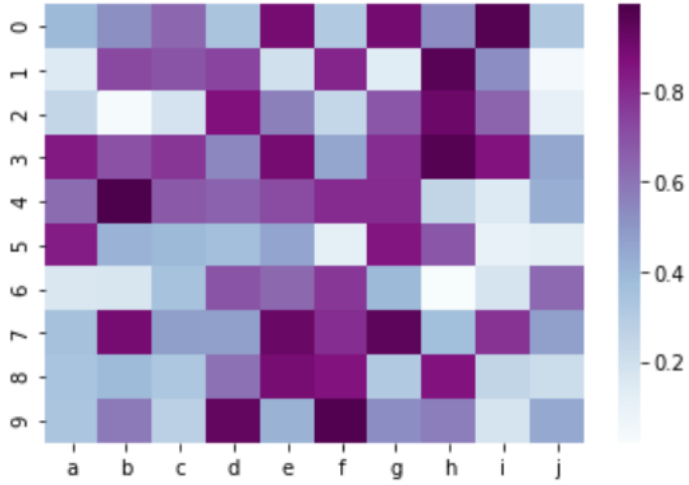
```
cmap="YlGnBu"
```

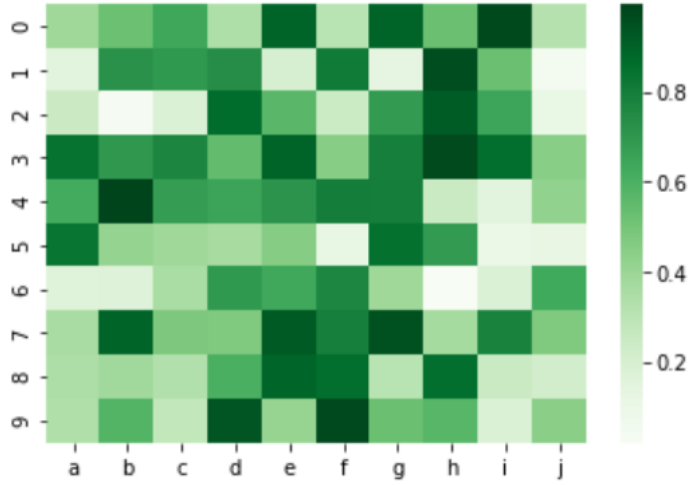





```
cmap="Blues"
```

```
cmap="BuPu"
```

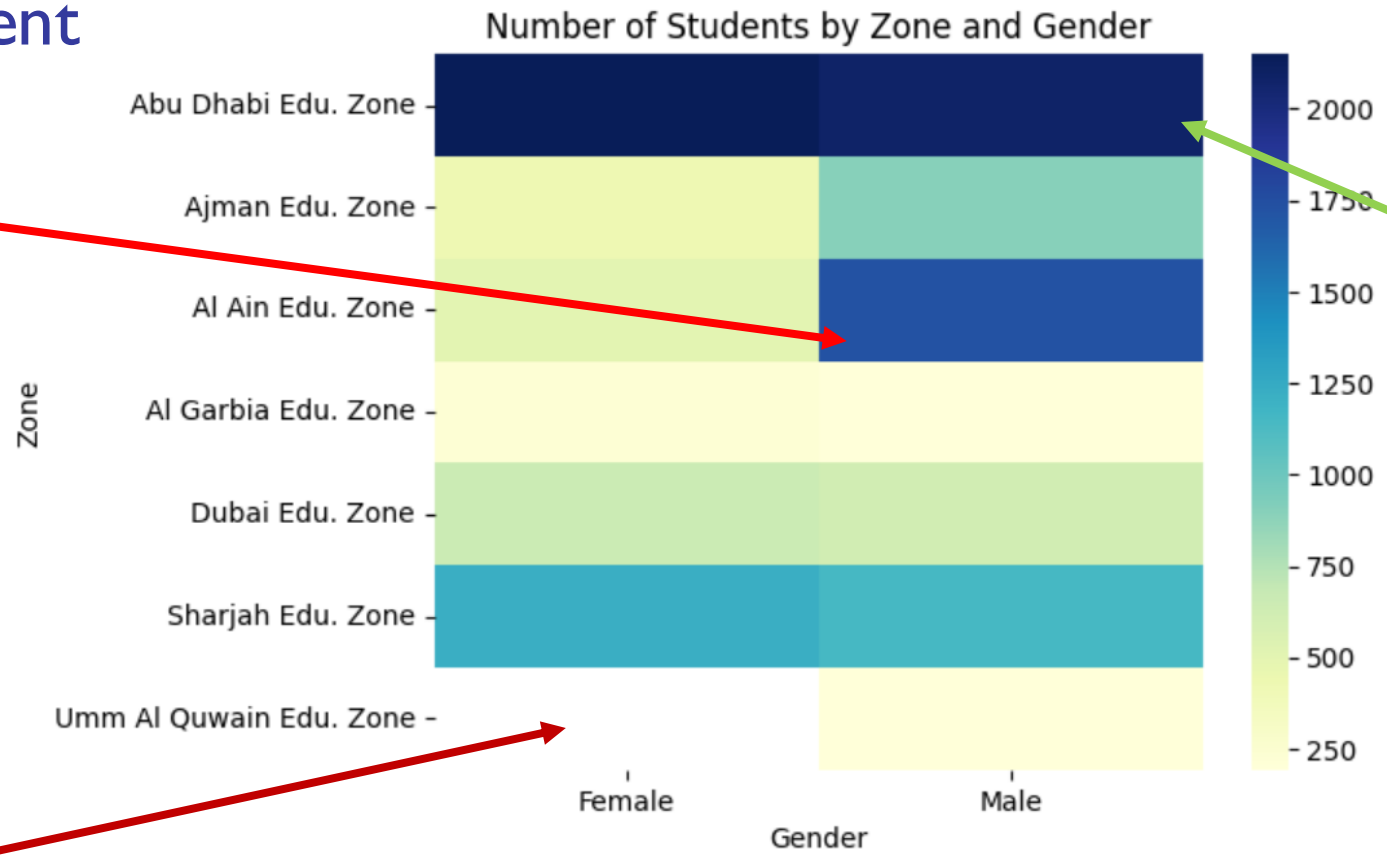





```
cmap="Greens"
```

Types of Charts/Graphs - Heat Map– Python

- Al Ain Male student population is around 1700



Why it is darker colour?

Why it is lighter colour?

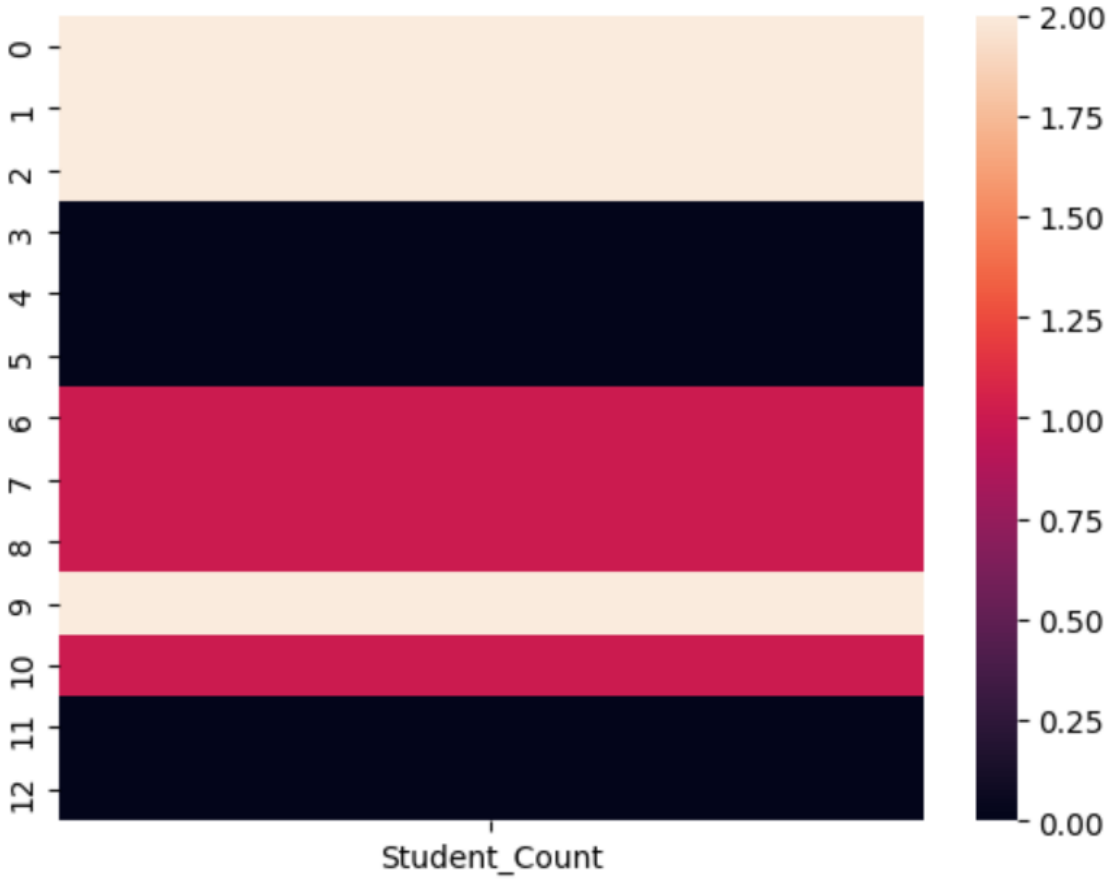
Types of Charts/Graphs - Heat Map – Python

- If your dataset consists of continues values, you can turn them into discrete numbers and use these discrete values in the heatmap.
- The following examples shows how to transform continues values(Student_Count) into 3 discrete values: 0, 1, and 2.

```
# libraries
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

# create data
df = pd.read_csv('Students.csv')

# make the values in dataset discrete
# the values will be cut into 3 discrete values: 0,1,2
df_q = pd.DataFrame()
for col in df:
  df_q['Student_Count'] = pd.to_numeric( pd.qcut(df['Student_Count'], 3, labels=list(range(3)))

# plot it
sns.heatmap(df_q)
plt.show()
```

Types of Charts/Graphs - Heat Map – Python

- Student population is divided into 3 segments and displayed on cherry red, black and apricot colour



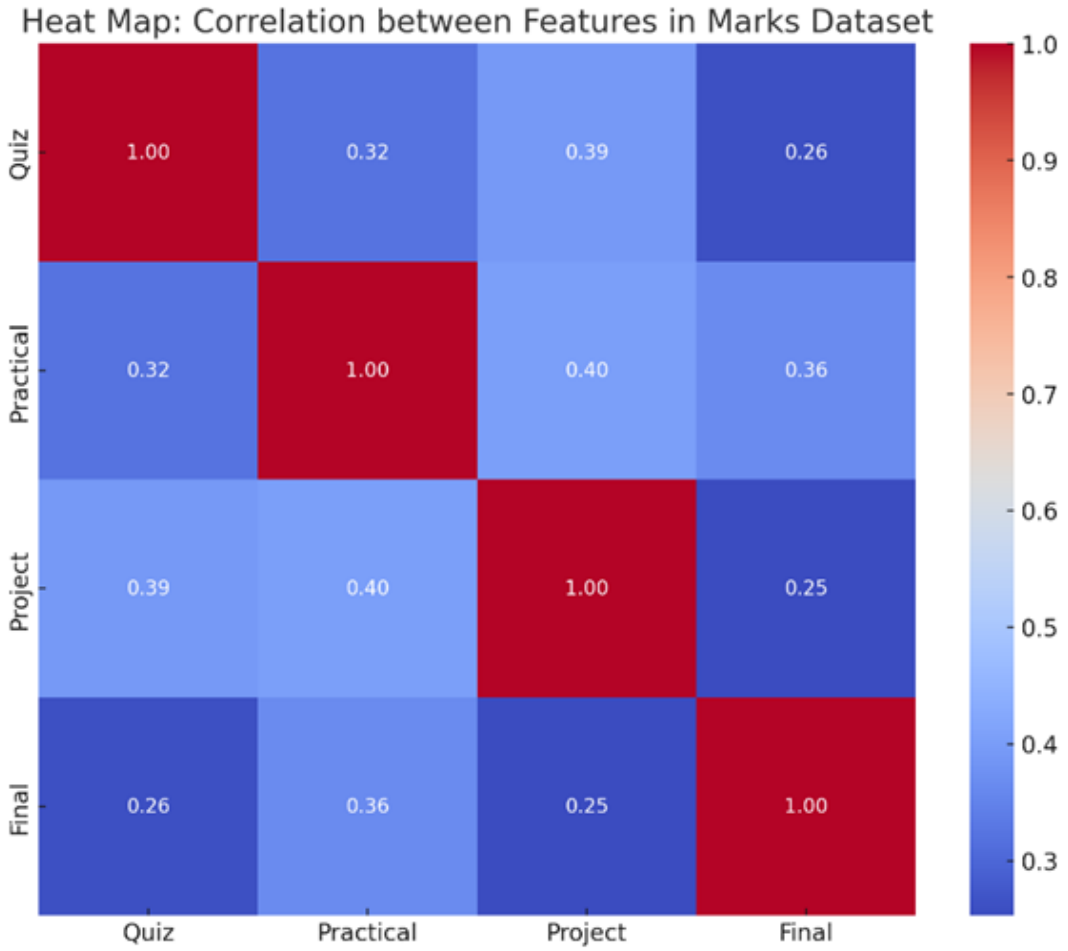
Quiz

A. Which two features have the strongest positive correlation in the dataset?

B. What does a dark red color on the heatmap represent?

C. What does the correlation coefficient of 0.32 between "Quiz" and "Practical" indicate?

D. Which two features have the weakest correlation in the dataset?



Quiz

E. Is there a strong correlation between "Quiz" and "Final" scores? Why or why not?

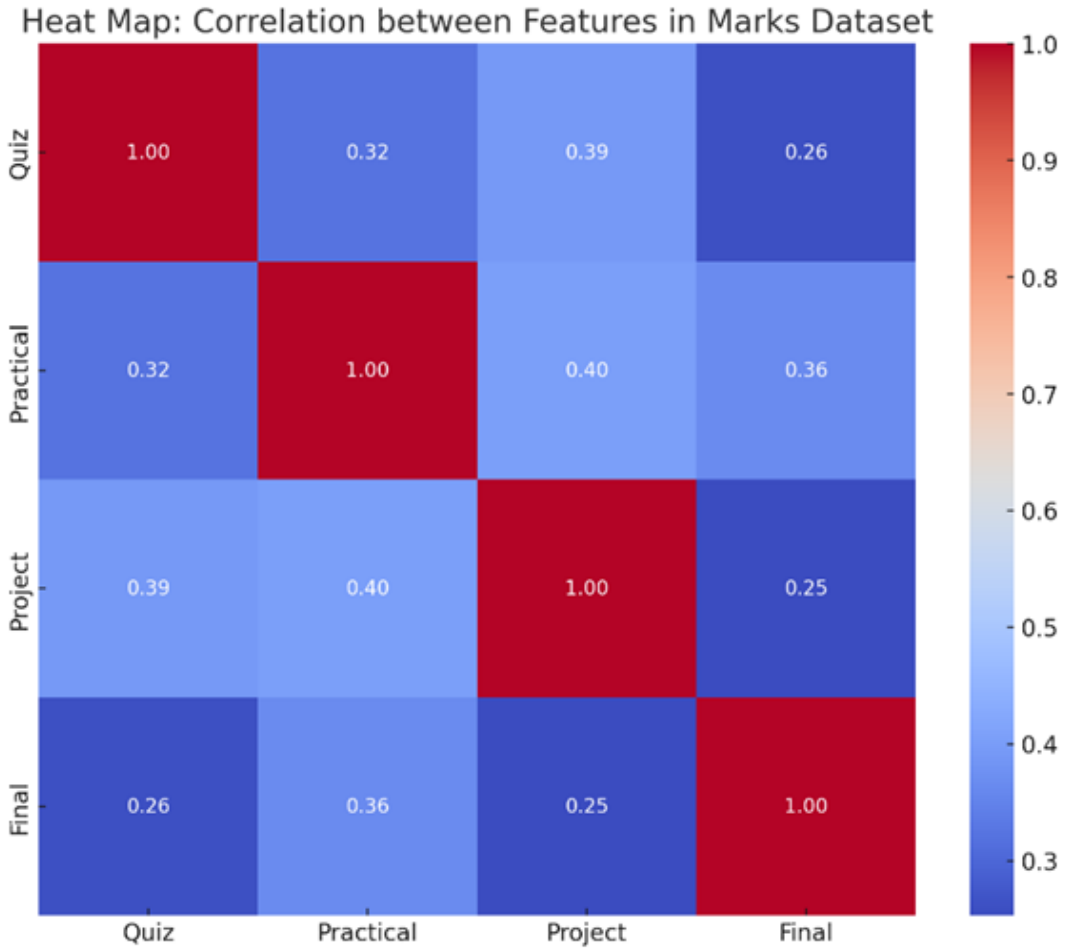
F. Based on the heatmap, which feature shows the highest overall correlation with "Final" scores?

G. If a student performs well in "Practical," can we predict their performance in "Project"? Justify your answer.

H. Why is there a lower correlation between "Quiz" and "Final" scores compared to "Practical" and "Final" scores?

I. Which feature has the most balanced correlations with other features in the dataset?

J. What can be inferred from the overall correlations in the dataset regarding the relationship between individual assessments and final scores?



Project

10. Visualize the dependent variable by the Graph/Chart of the following using Python Program: a. Scatter plot b. Box Plot c. Histogram d. Heat Map Hint: Use Matplot or Ski-learn library	3

Skill Yourself – Other Graph/Charts

Graph/Chart	Purpose	Python Libraries
Line Chart	Visualizes trends over time or continuous data.	matplotlib, seaborn, plotly
Bar Chart	Compares categorical data or discrete data points.	matplotlib, seaborn, plotly
Histogram	Displays the frequency distribution of a dataset.	matplotlib, seaborn, plotly
Box Plot (or Whisker Plot)	Shows the distribution of data and highlights outliers.	matplotlib, seaborn, plotly
Scatter Plot	Examines relationships or correlations between two variables.	matplotlib, seaborn, plotly
Heatmap	Visualizes correlations or relationships between variables in a matrix.	seaborn, matplotlib, plotly
Pie Chart	Displays proportional data as slices of a circle.	matplotlib, plotly
Violin Plot	Combines box plot and density plot for data distribution visualization.	seaborn, plotly
Density Plot	Visualizes the probability density of a dataset.	seaborn, matplotlib
Area Chart	Represents quantitative data and trends over time (like a line chart, but filled).	matplotlib, plotly
Bubble Chart	Visualizes three dimensions of data with size representing the third variable.	matplotlib, seaborn, plotly
Stacked Bar Chart	Compares parts of a whole across different categories.	matplotlib, seaborn, plotly
Grouped Bar Chart	Compares sub-categories within categories.	matplotlib, seaborn, plotly
Pair Plot	Visualizes pairwise relationships between variables in a dataset.	seaborn
Histogram with KDE	Combines a histogram with a Kernel Density Estimate curve.	seaborn, matplotlib
Treemap	Displays hierarchical data as nested rectangles.	plotly, squarify
Radar Chart	Compares multivariate data in a circular layout.	matplotlib, plotly
Chord Diagram	Shows relationships between different categories.	plotly, matplotlib
Network Graph	Visualizes relationships in networked data.	networkx, matplotlib, plotly
Sunburst Chart	Visualizes hierarchical data with a radial layout.	plotly
Candlestick Chart	Used for financial data to show open, close, high, and low values.	plotly, matplotlib
3D Scatter Plot	Explores relationships between three variables in 3D space.	matplotlib, plotly
Contour Plot	Visualizes 3D data in 2D using contours.	matplotlib, seaborn
Funnel Chart	Visualizes stages in a process or sales funnel.	plotly
Gantt Chart	Displays a project schedule or timeline.	plotly, matplotlib
Word Cloud	Visualizes text data with word frequency depicted as size.	wordcloud, matplotlib
Boxen Plot	Visualizes data distributions with emphasis on the tails.	seaborn
Facet Grid	Plots multiple subplots for visualizing relationships across subsets.	seaborn, matplotlib

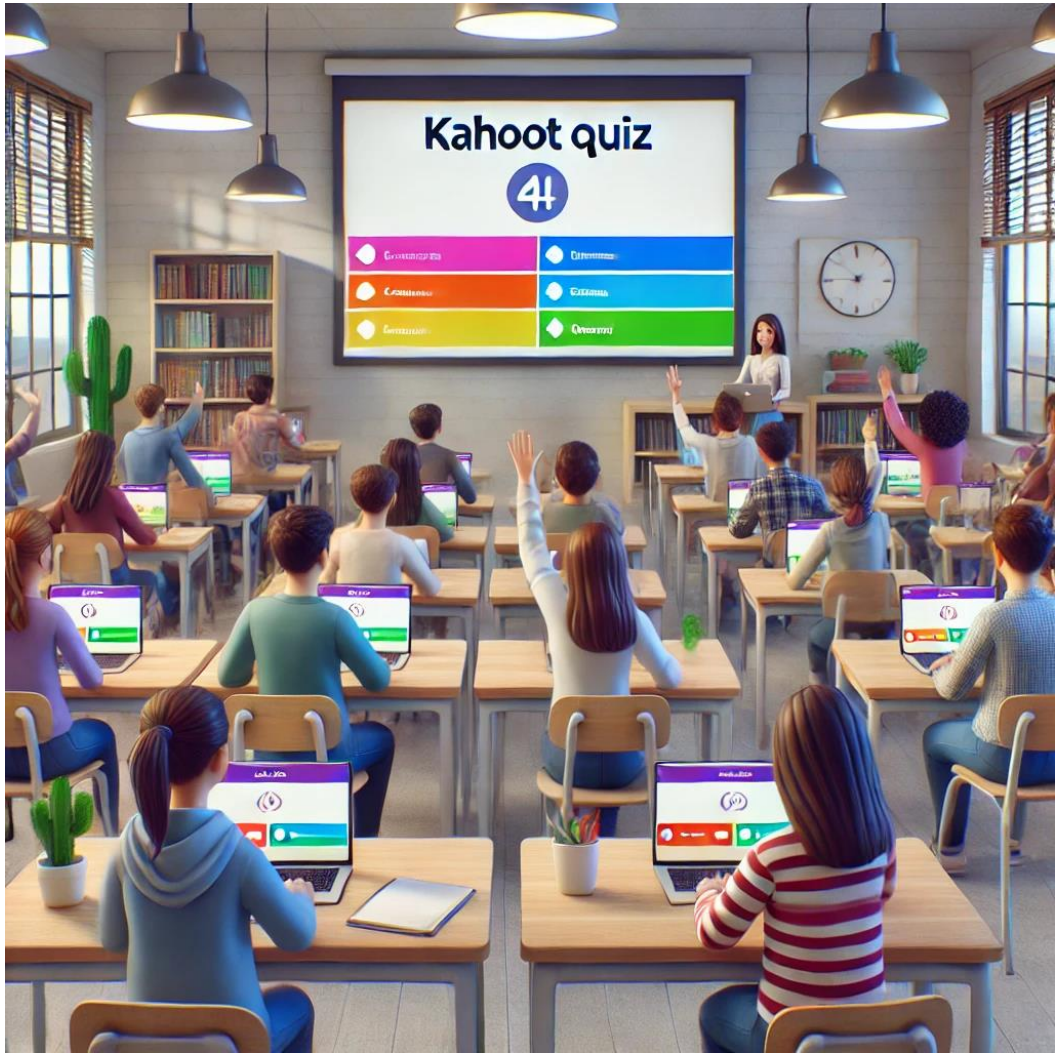
Conclusion

- Exploratory Data Analysis (EDA):
  - EDA involves various techniques to gain insights into the data, ensuring its correctness, identifying missing values, and detecting outliers.
  - The process includes steps like variable identification, univariate analysis, bivariate analysis, and handling missing values and outliers.
- Data Visualization:
  - Data visualization is the visual representation of data to facilitate the understanding of trends, patterns, and outliers.
  - Visualizations help present complex information in an accessible manner, making it easier for non-technical audiences to comprehend.

Conclusion

- Types of Charts and Graphs:
  - Scatter Plot: Shows the relationship between two variables and helps in identifying correlations.
  - Histogram: Displays the frequency distribution of a dataset, useful for understanding data distribution.
  - Box Plot: Provides a summary of data distribution, highlighting median, quartiles, and outliers.
  - Heat Map: Uses color gradients to represent data values, useful for identifying patterns and trends.
- Python Programming for Visualization:
  - Python libraries such as Matplotlib and Seaborn are powerful tools for creating various types of visualizations.
  - These libraries enable the creation of scatter plots, histograms, box plots, and heat maps, enhancing our ability to analyze and interpret data visually.

Kahoot Time!



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Thank You